Moon Formation by Triple Phase Transition in the Differentiating Proto-Earth

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Abstract. The Moon is there. This is a fact, not a hypothesis to be confirmed : our task is to trace its causes back. Just as our own existence attests that the improbable chain of conditions that made it possible actually took place, the existence of the Moon — with its mass, isotopic identity, and the angular momentum of the Earth–Moon system that we observe — imposes a precise set of conditions on the state of the Hadean proto-Earth. This theory does not seek to adjust free parameters to reproduce a desired outcome : it proceeds by *reconstruction*. The values we derive from it (radial density gradient, obliquity, angular momentum input) are not chosen ; they are constrained by a result that is, itself, certain. That a window of conditions is narrow is therefore not a fragility, but a retrodictive prediction about the early Earth.

The canonical giant-impact model faces growing geochemical difficulties : it does not naturally predict the near-isotopic identity of Earth and Moon, the crustal dichotomy, or the approximately 350 Ma delay in the Earth's dynamo. We propose an endogenous alternative, based on a necessary thermodynamic observation : any Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle, with accretion energy exceeding the melting energy by a factor of ≈ 121 (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The proto-Earth is thus a rapidly rotating magmatic body ($T_{\text{rot}} \approx 3.5$ h), without a stabilizing satellite, whose axis undergoes a large-amplitude free nutation in the interval $[40^\circ, 70^\circ]$ measured in its own co-rotating reference frame — a geometric internal oscillation of the fluid body (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014).

A unique engine — the progressive segregation of metallic iron-nickel toward the forming core — drives three coupled transitions. *(i)* A rheological transition : the magma acquires a Bingham-Herschel yield stress (Bingham, 1922; Herschel & Bulkley, 1926) and organizes a Coherent Magmatic Torus (CMT) in the intertropical belt $|\phi| < 30^\circ$. *(ii)* A mechanical transition : the CMT undergoes $N = 2$–3 episodes of cohesive hypersonic ejection. The ejection proceeds from a **deterministic dynamical instability** — a parcel of the torus leaves it when the Solberg-Høiland stability coefficient $\lambda(U)$ becomes positive. The threshold is reached cyclically : rise in angular momentum, ejection of a cohesive magmatic tongue at the nutation peaks, loss of momentum, recharge. Each ejection shifts the rotation axis toward

a new obliquity, and therefore the active belt itself : each successive CMT is geometrically distinct, which underlies the observed crustal dichotomy. *(iii)* A magnetic transition : the dynamo starts approximately 350 Ma after the end of the ejections, in agreement with Jack Hills zircons (Tarduno et al., 2025).

With an ejected mass of 2.2–4.0 $\times 10^{22}$ kg per episode and a capture efficiency of ≈ 0.70, the lunar mass is reconstructed without an external impactor. Isotopic identity and metallic iron depletion (reflected in a small lunar core) follow as direct mechanical consequences. The angular momentum budget, however, requires a post-formation external sink of approximately 3×10^{34} J s ; the evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) — which precisely requires the high obliquity predicted by the TPT — provides a quantified candidate, whose validation remains required (Limitation L15, Section 18). The distinction between metallic iron and oxidized iron is explicitly addressed in Section 14.4.

The central prediction — key to falsifiability — is the existence of one or more seismic interfaces between 200 and 530 km depth, with impedance contrast $|R| \in [0.01, 0.04]$, testable imminently by **Chang'e 7** (South Pole, August 2026), and subsequently by **FSS**, **LEMS** and **Artemis III** (2028–2029). Preliminary support comes from Chang'e-6 samples (Yue et al., 2026; Xu et al., 2025) : norites dated to 4247 ± 5 Ma and high Fe/Mn ratios in deep olivines, consistent with predictions P3 and P6. This retrodictive framework establishes coherence and physical plausibility of the scenario ; its distinction from the giant impact rests entirely on these observational predictions. If they are refuted, the theory must be refuted as well — and we wish, in that case, to one day learn what actually built the Moon.

Interactive simulation : https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html

Keywords : Moon origin · Triple Phase Transition · chaotic axial oscillation · Hadean proto-Earth · Bingham-Herschel · Coherent Magmatic Torus · generalized instability coefficient · parametric elliptical instability · seismic interface · Hadean dynamo · falsifiable predictions · South Pole–Aitken · Chang'e-6.

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1 Introduction : why a new theory of lunar formation ?

1.1 The problem of the giant impact

The canonical giant-impact model (Cameron & Ward, 1976; Hartmann & Davis, 1975) has dominated planetary science for nearly fifty years. In this scenario, a Mars-sized body (Theia) struck the proto-Earth, and the resulting debris disk accreted to form the Moon. This model explains several first-order characteristics : the angular momentum of the Earth–Moon system, the relatively modest size of the lunar core, and the absence of a massive atmosphere.

However, over the past decade, growing geochemical and chronological difficulties have eroded confidence in the giant-impact scenario as a complete explanation :

1. **Earth–Moon isotopic identity** : oxygen, titanium, tungsten, and chromium isotopic compositions are nearly identical between Earth and Moon to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The giant-impact model requires either that Theia had an isotopic composition identical to that of Earth (an improbable coincidence at $< 1\%$), or complete post-impact isotopic mixing (Canup, 2012), which is not naturally predicted by hydrodynamic simulations.
2. **Small lunar core** : the Moon has a metallic core of radius ≈ 330 km, significantly smaller than Earth's core relative to the host planet's mass (O'Neill, 1991; Jones & Drake, 1986). The giant impact explains this metallic iron depletion by assuming Theia was pre-differentiated (with a metallic core that accreted to Earth), but this adds a parameter.
3. **Crustal dichotomy** : the lunar farside is thicker (≈ 54 km) and older than the nearside (≈ 34 km) (Wieczorek et al., 2013). The giant impact does not predict this asymmetry.
4. **Delay in Earth's dynamo** : the Earth's magnetic field is detected in Jack Hills zircons at 4.2 Ga (Tarduno et al., 2025), i.e., approximately 350 Ma after the estimated formation of the Moon (4.55 Ga). The giant impact offers no explanation for this delay.

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in the formation of the Moon (Sossi et al., 2025). This conclusion motivates the exploration of alternatives based on the internal dynamics of the proto-Earth.

1.2 The Triple Phase Transition : an endogenous alternative

This work proposes that the Moon did not form by an external impact, but by an endogenous process within the proto-Earth itself during its differentiation. The central thesis is as follows.

Hypothesis H0 (central thesis). The Moon formed through a series of *three coupled transitions* — rheological, mechanical and magnetic — triggered by the unique engine of Fe-Ni segregation toward the forming core. The result is a Moon built layer by layer in two to three episodes of cohesive magmatic ejection, each layer archiving the state of Hadean differentiation at the time of its accretion.

The mechanism rests on a necessary thermodynamic observation : any Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The accretion energy exceeds the total melting energy by a factor of approximately 121. The proto-Earth is therefore a rapidly rotating magmatic body ($T_{\rm rot} \approx 3.5$ h) without a stabilizing satellite.

In this rapidly rotating fluid body, the rotation axis undergoes a large-amplitude free nutation in the interval $[40°, 70°]$ in the co-rotating reference frame (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014) — not an astronomical obliquity, but an internal oscillation of the fluid body. This oblique inclination is the geometric condition that allows the radial Coriolis force ($f_{\rm rad} = 2\Omega U \sin \varepsilon$) to destabilize the flow rather than confine it.

1.3 Organization of the manuscript

The manuscript is organized as follows. Section 2 presents the Hadean physical context (rotation, obliquity, silicate atmosphere). Section 5 presents the intertropical belt as the ejection site. Section 6 addresses the Taylor-Proudman objection. Section 7 inventories the forces. Section 8 explicitly classifies all hypotheses. Section 9 presents the three coupled transitions (rheological, mechanical, magnetic) with the complete resolution of the instability coefficient. Section 10 presents the mass budget. Section 11 presents angular momentum transport. Section 12 gives the chronology of 3 to 4 weeks. Section 13 presents the internal stratification. Section 14 discusses observational constraints and existing evidence. Section 15 presents the falsifiable predictions. Section 16 presents the validation windows. Section 17 discusses the main objections. Section 18 presents the limitations. Section 19 concludes.

2 The Hadean inferno : physical context

Before delving into the physics of the Triple Phase Transition, it is essential to establish a structural fact, not merely a descriptive one, about the framework in which it operates : no stationary state is physically accessible there. This is not a turbulent period among others in Solar System history — it is a regime where every process that could, in isolation, bring the system back toward equilibrium is itself perturbed, on a timescale shorter than its own relaxation time, by at least one other independent process. It is this structural absence of rest, and not an inventory of individual circumstances, that this section aims to establish : it directly conditions the reading of Section 3 (the axial oscillation is never damped for lack of a stabilizer) and, later, of Section 9.2 (each ejection episode occurs not on a system at rest, but on a system already and continuously out of equilibrium).

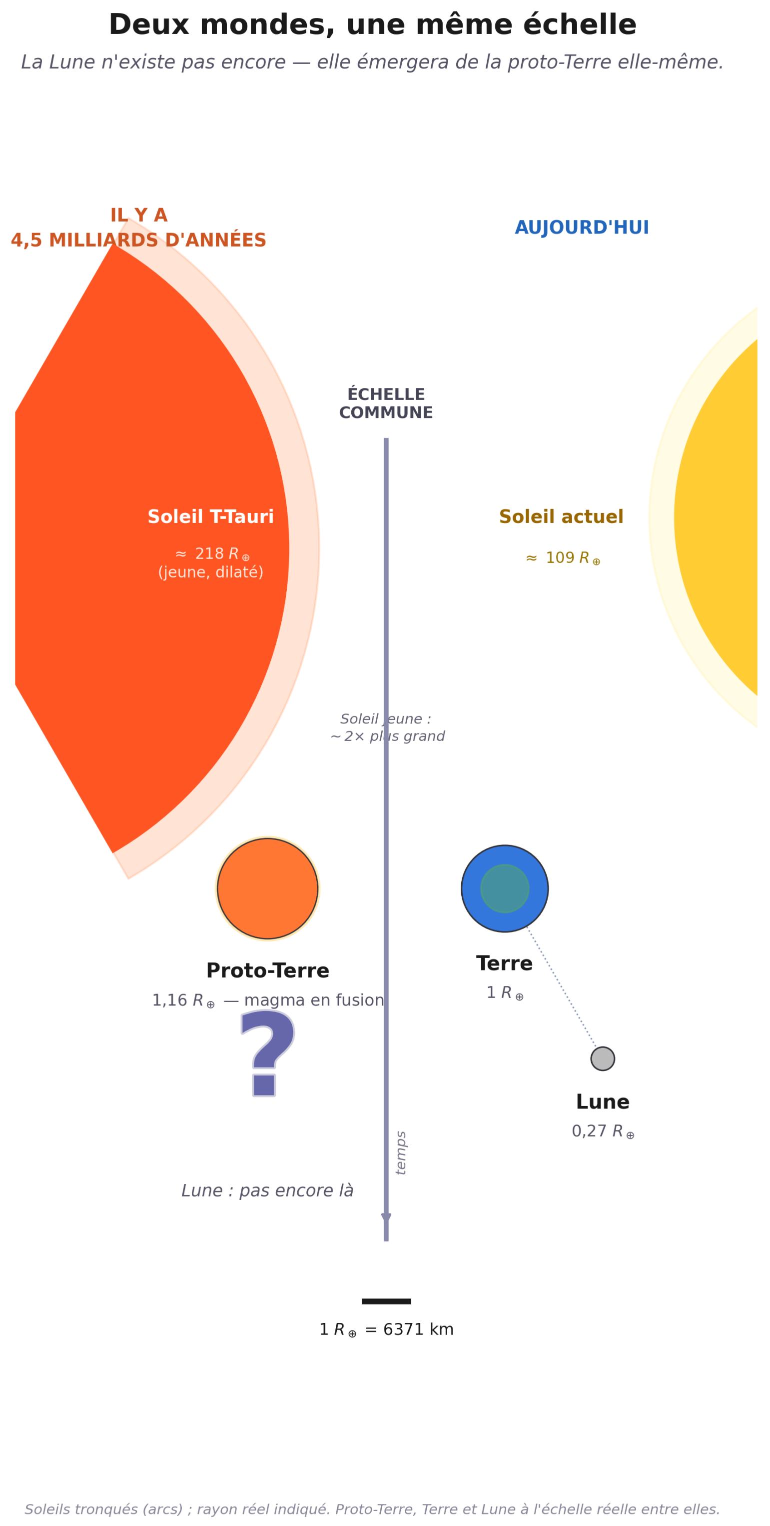

FIGURE 1 – Two worlds, same scale. Left, 4.5 billion years ago : the young, inflated T-Tauri Sun ($\approx 218\, R_\oplus$), and the molten proto-Earth ($1.16\, R_\oplus$) — the Moon does not yet exist ; it will emerge from the proto-Earth itself. Right, today : the present Sun ($\approx 109\, R_\oplus$), the Earth ($1\, R_\oplus$) and the Moon ($0.27\, R_\oplus$). Suns truncated (arcs), actual radius indicated ; proto-Earth, Earth and Moon at true relative scale.

2.1 The dynamical context of terminal accretion

The Hadean period (4.568–4.4 Ga) superimposes a set of physical processes whose characteristic timescales overlap without any one dominating long enough to impose a stable regime :

— **Continuous planetesimal accretion** : the protoplanetary disk remains populated by planetesimals and embryos, with each major impact occurring every 10^3–10^5 yr (Agnor et al., 1999; Kokubo & Genda, 2010) — an interval much shorter than the viscous relaxation timescale of the fluid body (Section 3), so that the effect of one impact never has time to fade before the next.

— **Rapid rotation** : the proto-Earth rotates with a period $\approx$ 3.5 h, a regime in which Coriolis and centrifugal effects dominate internal dynamics and amplify perturbations rather than damping them.

— **Absence of a tidal stabilizer** : without the Moon, obliquity is free to evolve over a broad chaotic range, with no restoring mechanism toward an equilibrium value.

— **A young, hyperactive Sun** : T-Tauri stars produce flares and coronal mass ejections (CMEs) at frequencies that can reach 0.2–0.4 events per day for superflares associated with confirmed mass ejections, and up to more than one event per day for white-light flares alone, based on direct observations of young solar-type stars ($\sim$ 30–125 Ma) (Namekata et al., 2026), compared to about one superflare per few thousand years for the present Sun (Feigelson & Montmerle, 1999; Airapetian et al., 2016) — i.e., a stochastic forcing with a characteristic daily timescale, whereas planetesimal impacts are millennial and the free nutation is also of the order of a day (Section 3) : these three clocks never synchronize, they telescope.

— **Short-lived radioactivity** : the isotopes ^{26}Al and ^{60}Fe are still active, injecting heat into the mantle continuously (Urey, 1955; Huss et al., 2009), with no thermal quiescence phase.

— **Dense silicate atmosphere** : a rock-vapor envelope at $T \sim 2,000$–$4,000$ K and $P \sim 10^5$–10^7 Pa constitutes a thermal blanket and an agent of lateral confinement (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), but does not itself stabilize anything : it extends the window during which the preceding processes can continue to act.

What matters is not the list of these processes, but their simultaneity. Taken separately, each could in principle average over time and allow a quasi-stationary behavior to emerge ; taken together, at frequencies that are commensurable on no common timescale, they structurally prevent such a regime from establishing itself. The T-Tauri Sun, although it provides one of the most frequent forcings, is therefore not the cause of Hadean chaos : it is but one contributor among others, whose addition makes any equilibrium inaccessible. It is in this already and continuously out-of-equilibrium context that the ejection episodes themselves will occur (Section 9.2) : not as isolated perturbations striking an otherwise calm system, but as additional shocks added to a system that, at no time during its active history, has truly rested.

2.2 The silicate atmosphere : thermal blanket and lateral confinement

At $t = 0$, the proto-Earth is enveloped by a dense silicate atmosphere — rock vapor, gaseous SiO, Mg and Fe — at temperatures of 2,000 to 4,000 K and pressures of 10^5–10^7 Pa. This cocoon plays two critical roles in the TPT.

First role : thermal insulation. The silicate atmosphere has high infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). Without this atmosphere, the proto-Earth would have cooled in $\sim 10^4$ yr. With it, the melting window extends to 30–50 Ma — exactly the timescale of core formation (Kleine et al., 2002; Yin et al., 2002). This is not an ad hoc hypothesis : it is the physical consequence of a rock-vapor atmosphere at $2,000$–$4,000$ K. The same mechanism operates in magma ocean models (Elkins-Tanton, 2008; Lebrun et al., 2013).

Second role : lateral confinement. The confining pressure ($P \approx 10^5$–10^7 Pa) opposes the lateral expansion of the CMT flow during ejection. It ensures the cohesion of the ejected sheet. The high sound speed in this torrid atmosphere ($c_{\text{son}} \approx 1,500$ m s^{-1} at $T_{\text{atm}} \approx 3,000$ K) reduces the effective Mach number of the flow, improving cohesion compared to ejection into vacuum.

Without this atmosphere, the melting window would close in $\sim 10^4$ yr. With it, the window extends to 30–50 Ma (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013), which corresponds to the timescale of core formation (Kleine et al., 2002; Yin et al., 2002).

2.3 Continuous planetesimal bombardment

The protoplanetary disk remains populated by planetesimals and embryos during the active TPT window (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2001). Each impact : (1) reinjects thermal energy into the mantle, sustaining melting ; (2) generates a spherical pressure wave propagating through the entire melt pool, perturbing $U(r)$; (3) delivers an impulsive torque on the spin vector, resetting the axial oscillation.

2.4 Short-lived radioactivity : internal heat source

The isotopes ^{26}Al (half-life 0.72 Ma) and ^{60}Fe (half-life 2.6 Ma) are still active during the first million years (Urey, 1955; Huss et al., 2009), injecting approximately 3×10^{29} J of additional heat into the mantle and contributing to sustained melting throughout the TPT window.

3 Hadean obliquity $\varepsilon \in [40°, 70°]$: a fundamental geometric clarification

The interval $\varepsilon \in [40°, 70°]$ is not a free parameter. It is constrained by five independent and convergent physical arguments. Before enumerating them, a fundamental geometric clarification is necessary.

Geometric clarification. In this work, ε denotes the instantaneous angle between the rotation vector $\mathbf{\Omega}(t)$ and the principal inertia axis of the Maclaurin spheroid, **defined in the body's co-rotating reference frame**. It is *not* the astronomical obliquity relative to the ecliptic plane : this concept has no physical meaning for the Hadean proto-Earth, which possesses neither a fixed reference ecliptic nor a Moon. It is a purely internal geometric angle describing the *oscillation* — the free nutation — of the rotation axis relative to the body's own shape.

Definition of the intertropical belt. The Hadean intertropical belt $|\phi| < 30°$ is defined perpendicular to the *instantaneous* rotation axis $\mathbf{\Omega}(t)$, and not relative to a fixed geographic reference frame. The plane $\phi = 0$ is thus the *instantaneous dynamical equator* of the body, which oscillates with the free nutation of $\mathbf{\Omega}(t)$ — it is not a fixed geographic equator. The coordinate ϕ is the oblique latitude measured from this instantaneous equator. It is the plane where the radial Coriolis force $f_{\text{rad}} = 2\Omega U \sin \varepsilon$ is maximal and where the effective gravity $g_{\text{eff}}(\phi)$ is minimal (Section 4.4). The belt is therefore a dynamical structure linked to the instantaneous equator, not a fixed latitude.

It follows directly from the absence of any external reference plane in the Solar System at that time — neither ecliptic, nor stable lunar orbit, nor fixed solar axis — that the very notion of an "intertropical" belt is not geographic but purely relative, defined by this internal axis rather than by a solar or ecliptic reference frame that does not yet exist. Throughout this work, terms evoking ordinary planetary geography ("intertropical", "equator", "latitude") must be read as a formal analogy with the geometry of a rotating spheroid, and not as a reference to an astronomy that had not yet occurred at that time.

Argument 1 — Absence of a tidal stabilizer (fundamental). The proto-Earth has no Moon. Without a massive satellite, there is no lunar tidal torque to damp the precession of the spin axis and keep it near a stable equilibrium. This argument is logically prior to all others : it is precisely the formation of the Moon that we seek to explain, so invoking it as a stabilizer would be circular. The absence of a tidal stabilizer is therefore a necessary, not contingent, condition of the Hadean proto-Earth.

Argument 2 — Laskar et al. (1993) : a chaotic zone without a satellite. Laskar et al. (1993a) showed that without its Moon, Earth's obliquity would evolve chaotically over a range extending from nearly 0° to approximately 85°. Although this result strictly applies to a present-day Earth rotating in about 24 h, it establishes the general principle : *the Moon is the stabilizer, and without it, large obliquity excursions are the norm*. For the Hadean proto-Earth, this principle applies *a fortiori*.

Argument 3 — Rapid rotation at $T_{\mathrm{rot}} = 3.5$ h. The precession constant satisfies $\alpha_{\mathrm{prec}} \propto \omega$. At $T_{\mathrm{rot}} = 3.5$ h, the precession frequency is approximately $7\times$ higher than at 24 h, pushing out of reach the spin-orbit secular resonances identified by Laskar et al. (1993a). Li & Batygin (2014), working on a Moonless Earth in rapid rotation, found that obliquity remains chaotic but *diffuses slowly* over a broad range. Rapid rotation does not stabilize the axis — it merely changes the nature of chaos, from resonant to diffusive.

Argument 4 — Continuous bombardment and stochastic chaos. Each major impact during terminal accretion (Agnor et al., 1999) abruptly resets the spin vector and the shape of the fluid body. For a low-viscosity fluid body ($\eta \sim 10^{1\text{–}3}$ Pa s), the viscous relaxation time toward axial alignment, of the form $\tau \sim \eta/(\rho g R_{\mathrm{eq,proto}})$ (Goldreich & Mitchell, 2010), is extremely short ($\ll 1$ yr) compared to the interval between major impacts (10^3–10^5 yr) : $\tau_{\mathrm{relax}} \ll \tau_{\mathrm{pert}}$. The system therefore realigns between each impact and spends more than 99.9% of the time aligned. Obliquity is thus not "continuously maintained" in the sense of a permanent upkeep of $\varepsilon > 45°$.

What impacts do ensure, however, is that the system never reaches a stationary state. Each impact resets the initial conditions of the free nutation, and the continuous stochastic forcing (CMEs, bombardment, late accretion) is distributed on timescales that are not commensurable. Obliquity is therefore not stabilized ; it is permanently *forced* toward new values. The interval $\varepsilon \in [40°, 70°]$ is not maintained by a slow relaxation mechanism, but by the absence of a stabilizer and the persistence of stochastic forcing that prevents the system from freezing. It is this absence of rest — and not an active maintenance — that makes large obliquity excursions accessible.

Argument 5 — T-Tauri CME torques. As described in Section 7.4, the impulsive torques of CMEs act on timescales of days to weeks, comparable to τ_{pert}, and constitute a continuous stochastic source of obliquity perturbation, in addition to impacts (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

The interval $\varepsilon \in [40°, 70°]$ is not an ad hoc hypothesis. It is the predictable consequence of the dynamics of a rapidly rotating fluid body, devoid of a tidal stabilizer, subjected to continuous stochastic perturbations. The simulations of Laskar et al. (1993a) for a Moonless Earth, and of Li & Batygin (2014) for a Moonless Earth in rapid rotation, converge on this interval — the same as that adopted by Laskar et al. (1993a) and Touma & Wisdom (1993). In this oblique geometry, the Coriolis force decomposes into radial and horizontal components (Section 7.3). The radial component $f_{\mathrm{rad}} = 2\Omega \sin\varepsilon \cdot U$ is maximal for $\varepsilon \in [40°, 70°]$.

Section synthesis

Five convergent arguments support $\varepsilon \in [40^\circ, 70^\circ]$. The absence of a tidal stabilizer is logically necessary. Arguments 2 to 5 provide independent physical justification for a large and sustained misalignment. The interval $[40^\circ, 70^\circ]$ is adopted as the physically motivated Hadean interval ; the mechanism operates for any $\varepsilon > 45^\circ$ (Section 7.3). The intertropical belt $|\phi| < 30^\circ$ is, throughout this work, linked to the instantaneous dynamical equator $\mathbf{\Omega}(t)$, and not to a fixed geographic reference frame.

3.1 Relaxation time and rheological reinforcement

The viscous relaxation time toward axial alignment for an isolated fluid body, $\tau_{\text{relax}} \sim \eta/(\rho g R_{\text{eq,proto}})$ (Goldreich & Mitchell, 2010), is very short ($\ll 1$ yr) at low viscosity, as established in Section 3. The Bingham-Herschel rheology locally slows this relaxation. The rheological response time to a stress perturbation is :

$$\tau_{\text{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\text{typ}}} \approx 10^2\text{–}10^4\,\text{s} \quad (\tau_y \sim 10^2\text{–}10^4\,\text{Pa},\ \mu \sim 1\text{–}100\,\text{Pa s},\ \dot{\gamma}_{\text{typ}} \sim 1\,\text{s}^{-1})\text{(Giordano et al., 2008)}, \tag{1}$$

where μ is the dynamic viscosity and $\dot{\gamma}_{\text{typ}}$ the typical shear rate. This time remains much shorter than the interval between impacts τ_{pert}, which means that mass redistribution is locally frozen by the yield stress between perturbations, delaying local shape relaxation without affecting the conclusion of Argument 4 : it is the absence of rest, due to the frequency of perturbations and the lack of a stabilizer, that keeps the system out of equilibrium.

4 Initial state of the proto-Earth

4.1 Near-total melting : a necessary thermodynamic consequence

The formation of an Earth-mass planet releases a considerable amount of gravitational energy. In the homogeneous sphere approximation (Safronov, 1969), the total accretion energy is :

$$E_{\text{grav}} = \frac{3\,G\,M_\oplus^2}{5\,R_{\text{eq,proto}}} \approx 1.93 \times 10^{32}\,\text{J}, \tag{2}$$

where G is the gravitational constant, $M_\oplus$ the Earth's mass, and $R_{\text{eq,proto}} \approx 7.4$ Mm the equatorial radius of the undifferentiated proto-Earth.

Equation (2) is a thermodynamic bound, not a model of instantaneous formation. It calculates the *total* gravitational energy released during accretion, independent of its duration. This is the standard approach in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The manuscript explicitly cites Kleine et al. (2002); Yin et al. (2002) as evidence that accretion lasted ~ 30 Ma.

The energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni, 2014) :

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24}\,\text{kg} \times 4 \times 10^{5}\,\text{J}\,\text{kg}^{-1} \approx 1.6 \times 10^{30}\,\text{J}, \tag{3}$$

where M_{mantle} is the mass of the silicate mantle and L_{fus} the specific latent heat of fusion.

The resulting energy ratio is :

$$E_{\text{grav}}/E_{\text{fus}} \approx 121. \tag{4}$$

This precise value follows from the parameters adopted here ($R_{\text{eq,proto}} \approx 7.4$ Mm, $M_{\text{mantle}} \approx 4 \times 10^{24}$ kg, $L_{\text{fus}} \approx 4 \times 10^{5}\,\text{J}\,\text{kg}^{-1}$), explicitly stated above ; it is not a universal constant of Earth. Other equally defensible parameter choices in the literature — a radius close to Earth's present radius, a smaller mantle mass, or a higher latent heat — give different numerical values, typically between approximately 80 and 190 depending on the assumptions adopted. What remains robust across this range, and what constitutes the consensus result in the literature (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015), is the qualitative conclusion : the accretion energy exceeds the melting energy of the silicate mantle by more than two orders of magnitude, making the near-total melting of the proto-Earth a robust thermodynamic consequence rather than a result contingent on a particular choice of parameters.

Why melting persisted despite radiative cooling

Accretion lasted $\sim$ 30 Ma (Kleine et al., 2002; Yin et al., 2002). During this period, radiative cooling occurred. Four independent mechanisms explain why melting persisted despite this cooling.

Mechanism 1 — The silicate atmosphere as a thermal blanket. The dense silicate atmosphere ($P \approx 10^5$–10^7 Pa, $T \approx 2{,}000$–$4{,}000$ K) has high infrared opacity. It reduces surface radiative cooling by a factor of 10^4 to 10^5 (Elkins-Tanton, 2008; Lebrun et al., 2013; Hamano et al., 2013). The effective cooling time increases from $\sim 10^4$ yr to 30–50 Ma.

Mechanism 2 — Additional heat sources. Four heat sources prolong melting beyond the initial accretion energy :

— **Fe-Ni segregation heat :** $\approx 3 \times 10^{30}$ J (Flasar & Birch, 1973; Rubie et al., 2003).

— **Short-lived radioactivity :** ^{26}Al (half-life 0.72 Ma) and ^{60}Fe (half-life 2.6 Ma) inject $\approx 3 \times 10^{29}$ J (Urey, 1955; Huss et al., 2009).

— **Continuous planetesimal bombardment :** reinjection of thermal energy (Agnor et al., 1999; Kokubo & Genda, 2010).

— **T-Tauri irradiation :** maintains surface temperatures above 2,000 K (Feigelson & Montmerle, 1999; Airapetian et al., 2016).

Mechanism 3 — The TPT does not require melting of the entire mantle. The Triple Phase Transition requires melting only within the Hadean intertropical belt $|\phi| < 30^\circ$ (upper and middle mantle). The factor 121 thermodynamically guarantees this partial melting, even if the deep mantle was not entirely molten (Nomura et al., 2011).

Mechanism 4 — The active TPT window is 30–50 **Ma.** The TPT occurs *during* the 30–50 Ma window during which melting is sustained by the mechanisms above. Core formation over $\sim$ 30 Ma (Kleine et al., 2002; Yin et al., 2002) is subsequent to this window, or simultaneous with it — it is not a contradiction.

Near-total melting constitutes the obligatory starting point. The objection based on the accretion timescale of $\sim$ 30 Ma does not contradict the mechanism : the TPT occurs *within* the melting window sustained by the silicate atmosphere, additional heat sources, and continuous bombardment.

Hypothesis H1 — Total volumetric melting. At the onset of Fe-Ni segregation, the entire silicate mantle lies above the liquidus, with no stable internal solid interface during the active window 4.568–4.4 Ga. This is a self-gravitating fluid volume, not a magma ocean resting on a substrate : the dynamics, rheology, and response to perturbations are physically distinct.

Caveat : Nomura et al. (2011) suggest that the deep lower mantle may not have been entirely molten, due to the elevation of the solidus with pressure. The TPT requires melting only within the Hadean intertropical belt $|\phi| < 30^\circ$ (upper and middle mantle), which the factor 121 thermodynamically guarantees. The adiabatic temperature profile (Appendix A) confirms melting throughout the mantle.

4.2 Initial rotation period : $T_{\mathrm{rot}} \approx 3.5$ h

The initial rotation period of the proto-Earth is not a free parameter : it is directly constrained by the literature on terrestrial planet formation to $T_{\mathrm{rot}} \approx$ 3–5 h.

1. T-Tauri magnetic torque. The young, rapidly rotating Sun ($\approx$ 3 days, Bouvier et al. 2014) magnetically couples to the inner protoplanetary disk (Königl, 1991; Shu et al., 1994), transferring angular momentum to matter on the inner orbit and accelerating the proto-Earth.

2. N-body simulations of terrestrial planet formation. N-body simulations show that the final spin periods of Earth-mass planets lie in the range 3–5 h (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2013). Agnor et al. (1999) find that protoplanetary embryos rotate rapidly during the later stages of accretion, at their first collision, suggesting that the proto-Earth may have rotated rapidly before any subsequent major impact. Kokubo & Genda (2010), using N-body simulations of protoplanet formation under realistic accretion conditions, find final rotation periods of 3–5 h for Earth-mass bodies, consistent with Chambers (2013). The value

$T_{\mathrm{rot}} \approx 3.5$ h adopted here lies within this independently established range, and is of the same order of magnitude as the pre-impact rapid rotations considered in the giant-impact models of Ćuk & Stewart (2012) (2.3–2.7 h) and Canup (2012) (a pre-impact rotation of 3 h in one of the simulated cases) — although none of these specific values is identical to 3.5 h, and the TPT does not rest on the giant-impact hypothesis itself.

Why rapid rotation requires an acceleration mechanism. Dones & Tremaine (1993) show analytically that orderly accretion from a uniform or slowly varying disk of small bodies cannot produce a rotation as rapid as that of Earth or Mars, regardless of the value of the disk velocity dispersion : the prograde and retrograde contributions of individual planetesimals nearly cancel by symmetry. The rapid spin rates of terrestrial planets are thus most naturally explained by one or a few late collisions between large bodies in accretion — a requirement that the TPT satisfies through the continuous planetesimal bombardment of Section 2, independent of whether or not recourse is made to a single dominant impactor (as in the giant-impact hypothesis). Consistent with this, Kokubo & Ida (2007), using N-body simulations of the embryo fusion phase of accretion, find that planetary spin angular velocities follow a Maxwellian distribution with substantial dispersion : there is no single preferred rotation period predicted by accretion physics alone, but only a broad statistical range within which $T_{\mathrm{rot}} \approx 3.5$ h is a plausible value.

3. Ice-skater effect : Fe-Ni segregation. The migration of Fe-Ni toward the forming core reduces the moment of inertia as segregation proceeds. With $I_f / I_0 \approx 0.82$ (Rubie et al., 2015), this contributes an additional acceleration of about 18% to the rotation over the TPT window, in addition to the rotation speed already acquired by the proto-Earth during accretion. This effect is qualitative and contributive : it is not used here to derive $T_{\mathrm{rot}} \approx 3.5$ h from an assumed pre-segregation period, since no such value is independently constrained.

4. Distinction from the synestia regime. Lock & Stewart (2017) define the corotation limit (CoRoL) as the rotation speed at which a body's equatorial velocity equals the local Keplerian orbital velocity ; bodies exceeding this limit form a distended, partially vaporized structure they call a synestia, rather than a simple rigidly rotating body. For the proto-Earth parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm), the corotation limit corresponds to a period of about 1.8 h. At $T_{\mathrm{rot}} \approx 3.5$ h, Ω reaches only about 50% of this limit : the proto-Earth described by the TPT thus lies well within the regime of a coherent rigidly rotating body, and not a synestia.

4.3 Why rapid rotation : a dedicated subsection

The rapid rotation of the proto-Earth ($T_{\mathrm{rot}} \approx 3.5$ h) is not a decorative hypothesis. It is required by the physics of accretion and by the TPT mechanism itself. This subsection summarizes the five independent reasons that constrain this value.

1. **Orderly accretion is insufficient.** Dones & Tremaine (1993) show analytically that symmetric accretion from a disk of small bodies cannot produce rapid rotation : prograde and retrograde contributions nearly cancel. A rapid spin requires an acceleration mechanism late in accretion.
2. **N-body simulations impose 3–5 h.** Kokubo & Ida (2007) find that the angular velocities of terrestrial planets follow a Maxwellian distribution with substantial dispersion. 3.5 h is a plausible value within this range.
3. **Fe-Ni segregation accelerates rotation.** The migration of metallic iron toward the core reduces the moment of inertia by about 18% (Rubie et al., 2015), which accelerates rotation by the ice-skater effect.
4. **The ejection threshold requires rapid rotation.** The condition $\Omega > \Omega_{\mathrm{crit}}$ (Section 9.2) is only reachable for $T_{\mathrm{rot}} \lesssim 4$ h. A slower proto-Earth could never eject material.
5. **The proto-Earth is not a synestia.** The corotation limit (CoRoL) of Lock & Stewart (2017) is at ≈ 1.8 h for the proto-Earth parameters. At 3.5 h, Ω reaches only about 50% of this limit : the body remains a coherently rotating spheroid, not a distended structure.

On the initial rotation period

The value $T_{\mathrm{rot}} \approx 3.5$ h is directly constrained by the literature on terrestrial planet formation (Agnor et al. 1999; Kokubo & Genda 2010; Chambers 2013; 3–5 h), is of the same order of magnitude as the rapid rotations considered in giant-impact models (Ćuk & Stewart, 2012; Canup, 2012) without depending on the giant-impact hypothesis, and is consistent with the theoretical requirement that a rapid terrestrial spin requires a late accretion acceleration mechanism (Dones & Tremaine, 1993), within the broad statistical range found for planetary spin in N-body simulations (Kokubo & Ida, 2007). It is not derived from an assumed initial period via a contraction argument. The Coriolis compensation parameter b', a simple ratio between the critical ejection speed U_{crit} and the geostrophic equilibrium speed of the CMT, provides an independent cross-check : Fe-Ni segregation increases b', ejections decrease it, and the equilibrium point $b' = 1$ is a stable attractor that naturally selects $T_{\mathrm{rot}} \approx 3.5$ h and $\varepsilon \approx 55°$. The complete derivation is given in Appendix B.

4.4 Geometry of the proto-Earth : the Maclaurin spheroid

A rapidly rotating fluid mass adopts the shape of a Maclaurin spheroid (Chandrasekhar, 1969). The equatorial radius is $R_{\mathrm{eq,proto}} \approx 7.4$ Mm, combining reduced density (+4%), thermal expansion (+7.5%) and equatorial bulge (+7.5%). The dynamical flattening, given by $J_2 = q/[2\Phi(\bar{C}) - 3]$ where $q = \Omega^2 R^3_{\mathrm{eq,proto}}/(GM_\oplus)$ is the rotation parameter and $\bar{C}$ the normalized axial moment of inertia (Murray & Dermott, 1999), reduces to $J_2 \approx q/2$ for a strictly homogeneous body ($\bar{C} = 0.4$) :

$$J_2 \approx \frac{1}{2} \cdot \frac{\Omega^2 R^3_{\mathrm{eq,proto}}}{GM_\oplus} \approx 0.13, \tag{5}$$

a value consistent with the range $J_2 \sim 0.10$–0.13 expected for a body near homogeneity (Murray & Dermott, 1999), giving an equatorial bulge $\delta R \approx 600$–650 km. The effective gravity varies with latitude ϕ (measured from the instantaneous dynamical equator, as defined in Section 3) according to :

$$g_{\mathrm{eff}}(\phi) = g^{(0)}_{\mathrm{eff}} \left(1 + \tfrac{3}{2} J_2 \sin^2\phi\right) - \Omega^2\, r(\phi) \cos^2\phi, \tag{6}$$

where $r(\phi)$ is the local radius. The effective gravity is minimal within the belt $|\phi| < 30^\circ$: this is the geometric attractor of the CMT.

The free nutation (Euler) period, $\omega_E = \Omega(C - A)/A$ where C and A are the polar and equatorial moments of inertia of the spheroid (Murray & Dermott, 1999), with $(C - A)/A \equiv f \approx 0.075$ for a body near homogeneity at the dynamical flattening factor J_2 adopted here (Eq. 5), is :

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5}\,\mathrm{rad\,s^{-1}}, \qquad T_{\mathrm{nut}} \approx 46.7\,\mathrm{h}. \tag{7}$$

This frequency plays a central role in the parametric elliptical instability (Section 7.6).

Stability of the Maclaurin spheroid at this rotation rate. A self-gravitating fluid body in rapid rotation is not necessarily constrained to adopt the axisymmetric Maclaurin shape : beyond a critical rotation rate, the Maclaurin sequence bifurcates to the triaxial Jacobi sequence. The bifurcation criterion compares two characteristic timescales : the self-gravitation timescale, $t_g \sim 1/\sqrt{G\bar{\rho}}$, and the rotation period, $t_{\mathrm{rot}} = 2\pi/\Omega$; their ratio defines the dimensionless number $\Omega^2/(\pi G\bar{\rho})$. This criterion is derived analytically by Chandrasekhar (1969) (Chapter 4, complete demonstration ; see also Chapters 3 to 5 for Maclaurin spheroids and Chapters 6 and 7 for Jacobi ellipsoids and the bifurcation itself) as the point where the family of Jacobi ellipsoids branches off from the Maclaurin sequence — and not as an empirical approximation — and is presented in historical perspective by Jeans (1929), taken up in the stellar context by Tassoul (1978), the theory being formally identical for any self-gravitating fluid ; Lamb (1932) gives the classical hydrodynamic treatment, and Durand (1964) the ellipsoidal integrals on which Chandrasekhar's demonstration rests. This bifurcation occurs at $\Omega^2/(\pi G\bar{\rho}) \approx 0.374$, Maclaurin spheroids becoming truly unstable only beyond $\Omega^2/(\pi G\bar{\rho}) \approx 0.440$.

The ρ in this criterion is the uniform density of the homogeneous fluid in the theoretical model. Its application to a real planetary body substitutes for this uniform density the mean density $\bar{\rho} = M/(\frac{4}{3}\pi R^3)$, an approximation explicitly discussed by Chandrasekhar (1969) (Chapters 1 and 5) and Tassoul (1978) (Chapter 2). For the Hadean proto-Earth, $\bar{\rho}$ is not directly measurable : it depends on the degree of differentiation already achieved, which is itself not precisely known. Available estimates place the mean density of a homogeneous Earth of the same mass, before complete core segregation, around $4,100$–$4,500\ \mathrm{kg\,m^{-3}}$, compared to $5,510\ \mathrm{kg\,m^{-3}}$ for the fully differentiated present Earth (Rubie et al., 2015; Trønnes et al., 2019; Rubie et al., 2016; Hirose et al., 2021). These values themselves are derived from mineralogical models (chondritic or pyrolitic composition, equations of state) rather than from direct measurement, and thus constitute a modeling range rather than a physical constant.

For the proto-Earth parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm, mean density $\bar{\rho} \approx 4,300\ \mathrm{kg\,m^{-3}}$, the median value of this range, for a still weakly differentiated body), $T_{\mathrm{rot}} \approx 3.5$ h gives $\Omega^2/(\pi G\bar{\rho}) \approx 0.276$ — below the Jacobi bifurcation point (0.374), with a margin of about 26%. At this density, the bifurcation threshold itself corresponds to $\Omega_c = \sqrt{0.374\,\pi G\bar{\rho}} \approx 5.8 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, i.e., $T_{\mathrm{rot}} \approx 3.0$ h, and the strict dynamical instability threshold to $T_{\mathrm{rot}} \approx 2.8$ h. The axisymmetric Maclaurin geometry adopted throughout this work is therefore consistent with the TPT rotation rate, and does not require recourse to a triaxial or otherwise more complex equilibrium figure ; the available margin nevertheless remains sensitive to the exact value of $\bar{\rho}$ adopted, which is not known with precision better than the modeling range cited above.

5 The intertropical belt $|\phi| < 30^\circ$: the ejection site

A fundamental and often misunderstood point : the CMT is not an equatorial structure. It is a coherent zonal belt extending over $\pm 30^\circ$ of oblique latitude — about one third of the proto-Earth's surface. This extension results from three independent and convergent mechanisms.

As established in Section 3, the belt $|\phi| < 30^\circ$ is defined perpendicular to the instantaneous rotation axis $\boldsymbol{\Omega}(t)$, and not relative to a fixed geographic reference frame. It is the plane where the radial Coriolis force is maximal and where effective gravity is minimal.

Mechanism 1 — Maclaurin geometry. Effective gravity is minimal within the belt $|\phi| < 30^\circ$ (Eq. 6). Molten material accumulates there by centrifugal effects, independently of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

Mechanism 2 — Geostrophic inverse cascade. The Rossby number of the CMT, with $U_{\mathrm{turb}} \approx 0.8\ \mathrm{km\,s^{-1}}$ (Eq. 19), is :

$$\mathrm{Ro} = \frac{U_{\mathrm{turb}}}{2\,\Omega\,R_{\mathrm{eq,proto}}} \approx 0.108 \ll 1, \tag{8}$$

In this regime, turbulent energy is transferred to large scales (inverse cascade) (Rhines, 1975). The Rhines scale :

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}}, \qquad \beta = \frac{2\Omega\cos\varepsilon}{R_{\text{eq,proto}}}, \tag{9}$$

gives $L_\beta \approx 0.47\, R_\oplus$ — corresponding to the harmonic mode $\ell = 1, m = 1$: the largest coherent structure possible on a sphere. The CMT is the statistical attractor of this cascade (Rhines, 1975; Sukoriansky et al., 2007).

Mechanism 3 — The radial Coriolis force. $f_{\text{rad}} = 2\Omega U \sin\varepsilon$ concentrates the flow within the oblique intertropical belt and maintains it coherent there.

Section synthesis

These three mechanisms — Maclaurin geometry, geostrophic inverse cascade, and radial Coriolis force — are independent and converge on the same belt $|\phi| < 30^\circ$. Their simultaneous convergence constitutes an argument for the structural robustness of the theory.

6 The CMT as a purely toroidal flow : response to the Taylor-Proudman objection

Any geophysical fluid dynamicist confronted with the CMT will naturally raise the following objection : in a rapidly rotating fluid (Ro $\ll$ 1), the Taylor-Proudman theorem imposes $(\boldsymbol{\Omega} \cdot \nabla)\boldsymbol{u} = 0$, generating columnar structures parallel to $\boldsymbol{\Omega}$, rather than a single oblique belt $|\phi| < 30^\circ$. The objection is legitimate and deserves a precise response.

6.1 The poloidal/toroidal decomposition

Any velocity field within a spherical shell decomposes into a poloidal part (radial and meridional components) and a toroidal part (purely azimuthal component) :

$$\boldsymbol{u} = \boldsymbol{u}_{\text{pol}} + \boldsymbol{u}_{\text{tor}}, \qquad \boldsymbol{u}_{\text{tor}} = \nabla \times (\psi \hat{\boldsymbol{r}}). \tag{10}$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which carries vorticity along $\boldsymbol{\Omega}$) and produces the Taylor columns. It does not constrain the purely toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\phi = u_\phi(r, \phi)$), the Taylor-Proudman constraint is automatically satisfied :

$$(\boldsymbol{\Omega} \cdot \nabla) u_{\text{tor}} = 0 \tag{11}$$

for any toroidal field, without additional condition. The equilibrium is cyclostrophic, not geostrophic : the centrifugal pressure gradient balances the Coriolis force within the azimuthal plane.

6.2 Why curved columns do not extend beyond the active belt

In a spherical shell in oblique rotation ($\varepsilon \neq 0$), the Taylor columns are curved along the projected field lines of $\mathbf{\Omega}$. They tend to channel the flow away from its source — unless a cutoff mechanism stops them.

This is precisely the role of the yield stress τ_y. In a Bingham-Herschel fluid, flow propagates only in regions where the stress exceeds τ_y. As soon as the columns penetrate a region where $\tau < \tau_y$, they are truncated : the magma solidifies there (Frigaard & Nouar, 2017; Balmforth & Craster, 2001). The belt $|\phi| < 30°$ is precisely the region where effective gravity is minimal (Maclaurin) and where Coriolis stress is maximal ($\varepsilon \approx 55°$) : it is there, and only there, that $\tau > \tau_y$. The curved columns issuing from the active belt are thus naturally truncated at its boundaries.

Section synthesis

The CMT is a purely toroidal flow. The Taylor-Proudman objection does not apply. What remains to be demonstrated numerically — identified here as Limitation L1 (Section 18) — is whether a single torus forms within the belt $|\phi| < 30°$ rather than multiple alternating bands. This is a question for simulation, not a theoretical inconsistency.

7 Complete inventory of forces

All forces, projections and equations are expressed in the Hadean instantaneous co-rotating reference frame of the body, with ε defined as in Section 3.

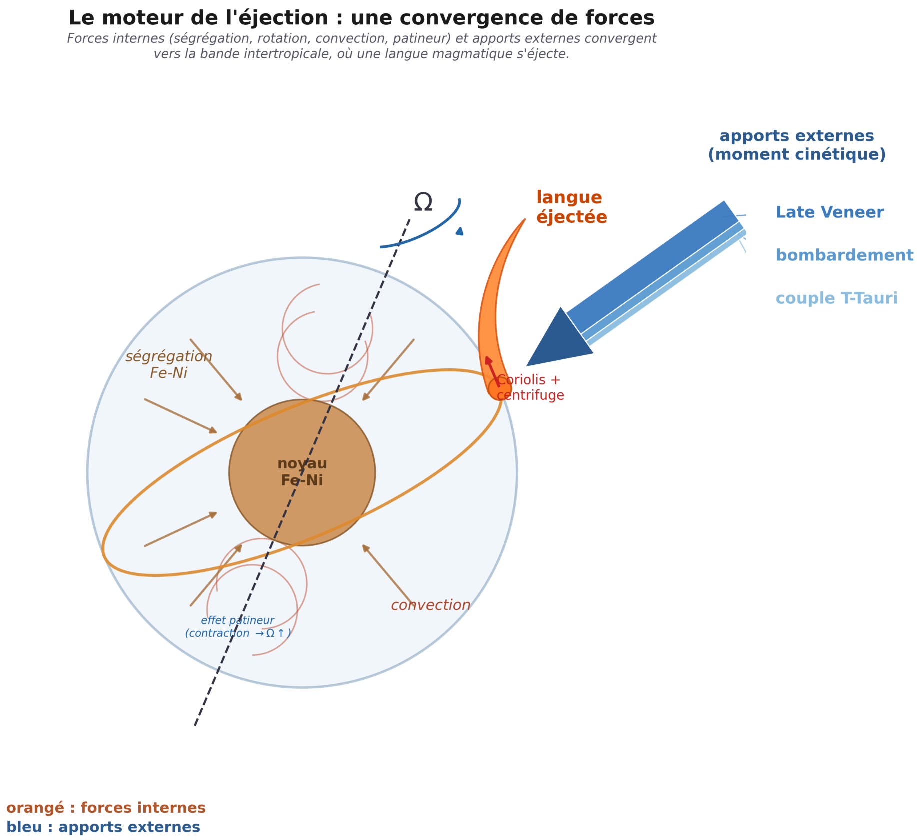

FIGURE 2 – The ejection engine : a convergence of forces. Internal forces (Fe-Ni segregation toward the core, rotation Ω, convection, ice-skater effect : contraction $\rightarrow \Omega \uparrow$) and external angular momentum inputs (Late Veneer, bombardment, T-Tauri torque) converge toward the intertropical belt, where a magmatic tongue is ejected under the combined effect of Coriolis and centrifugal forces. Orange : internal forces ; blue : external inputs.

7.1 Complete inventory of forces : summary table

The CMT is a toroidal flow within a rapidly rotating fluid body, subject to a set of forces. Table 1 summarizes all forces, their physical role and their mathematical expression.

TABLE 1 – Complete inventory of forces acting on the CMT.

Force	Role	Expression
Effective gravity	Radial confinement	$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)}(1 + \frac{3}{2}J_2 \sin^2 \phi) - \Omega^2 r(\phi) \cos^2 \phi$
Centrifugal force	Lateral ejection, flattening	$F_{\text{cent}} = \Omega^2 r(\phi) \cos^2 \phi$
Radial Coriolis	Destabilization, outward ejection	$F_{\text{cor,rad}} = 2\Omega U \sin \varepsilon$
Horizontal Coriolis	Lateral confinement	$F_{\text{cor,hor}} = 2\Omega U \cos \varepsilon$
Taylor-Proudman	Vertical columns, toroidal flow	$(\boldsymbol{\Omega} \cdot \nabla)\boldsymbol{u}_{\text{pol}} = 0$; $\boldsymbol{u}_{\text{tor}}$ unconstrained
Planetesimal bombardment	Stochastic perturbation, thermal reinjection	$\delta\boldsymbol{\Omega} = \sum_i \delta\boldsymbol{\Omega}_i$, $\delta U = \sum_i \delta U_i$
T-Tauri (CME)	Stochastic perturbation, thermal maintenance	$\boldsymbol{F}_{\text{CME}}(t) = \sum_j \boldsymbol{F}_{\text{CME},j}\, \delta(t - t_j)$
Solar tide	Low-frequency forcing on obliquity	$F_{\text{tide,sol}} \sim 10^{-4} g_0 \cos(\omega_{\text{sol}} t)$
Proto-lunar tide	Mathieu resonance (episodes 2+)	$F_{\text{tide,moon}}(t) = A_m^{(k)}(t) \cos(n_{\text{Moon}} t)$
Euler acceleration	Axial nutation	$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$
Atmospheric pressure	Lateral confinement	$P_{\text{atm}} \sim 10^5 - 10^7$ Pa
BH yield stress	Plasticity threshold	$\tau = \tau_y + k\dot{\gamma}^n$, $\tau > \tau_y$
Viscous stress	Dissipation	$\tau_{\text{visc}} = \eta \nabla^2 \boldsymbol{u}$

7.2 Oscillation as a direct source of force

Axial oscillation implies $\dot{\boldsymbol{\Omega}} \neq \boldsymbol{0}$. Poincaré (1910) showed that this generates an Euler acceleration on each fluid parcel :

$$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}, \tag{12}$$

where $\boldsymbol{r}$ is the position vector. The radial projection :

$$\boldsymbol{a}_{\text{Euler}} \cdot \hat{r} = |\dot{\Omega}|\, r(\phi)\, \sin(\varepsilon + \phi), \tag{13}$$

is maximal within the belt $|\phi| < 30°$ where $r(\phi)$ is largest.

7.3 Radial Coriolis force and geometric threshold

The Coriolis force on a fluid parcel moving at azimuthal speed U decomposes, in the oblique co-rotating frame, into :

$$f_{\mathrm{rad}} = 2\,\Omega\,\sin\varepsilon \cdot U \quad \text{(radial, outward, destabilizing),} \tag{14}$$
$$f_{\mathrm{horiz}} = 2\,\Omega\,\cos\varepsilon \cdot U \quad \text{(horizontal, confining),} \tag{15}$$

with the ratio :

$$\frac{f_{\mathrm{rad}}}{f_{\mathrm{horiz}}} = \tan\varepsilon. \tag{16}$$

Geometric threshold $\varepsilon = 45^\circ$ — necessary condition for radial ejection. For $\varepsilon > 45^\circ$: $\tan\varepsilon > 1$, the destabilizing radial component dominates the confining horizontal component. For $\varepsilon < 45^\circ$: the horizontal component dominates and the Taylor-Proudman constraint inhibits radial ejection. This threshold follows solely from Coriolis kinematics : no adjustable parameter is involved. At $\varepsilon = 55^\circ$: $f_{\mathrm{rad}}/f_{\mathrm{tot}} = \sin 55^\circ \approx 0.82$, i.e., 82% of the total Coriolis force is directed radially outward.

7.4 The T-Tauri Sun : catalyst and perturber

The Sun in its T-Tauri phase is not a driving force in the same sense as gravity or the Coriolis force — it does not act continuously on each fluid parcel of the CMT. It nevertheless plays three essential roles within the TPT window.

(1) Rheological modulator. T-Tauri UV/X-ray radiation maintains T_{surf} in the range $1,800$–$4,200$ K (Stefan-Boltzmann law, albedo $A_0 \approx 0.1$), which keeps the yield stress τ_y of the silicate magma in the range 10^2–10^4 Pa (Giordano et al., 2008) — the optimal window for CMT cohesion. Below $T^* \approx 1,800$ K, τ_y rises to 5.5×10^3–8.0×10^4 Pa (Eq. 25) and cohesive ejection becomes progressively more difficult. The T-Tauri Sun thus acts as a *guardian of the rheological window* via surface radiative forcing transmitted to the mantle by conduction through the silicate atmosphere.

(2) Source of stochastic perturbations. CMEs (10^{27}–10^{29} J per event, several per day) and radiativepulsations provide a continuous source of perturbation.

(3) Perturber of obliquity. CMEs acting on the proto-atmosphere and magnetosphere deliver impulsive torques on the spin vector, contributing to the maintenance of axial oscillation $\varepsilon \in [40^\circ, 70^\circ]$ (Section 3). The T-Tauri phase also provides an independent shutdown mechanism : once the Sun reaches the main sequence (≈ 4.4 Ga), irradiation declines, τ_y increases, and the TPT window closes.

The T-Tauri Sun is therefore not the *engine* of the mechanism — gravity, the Coriolis force, the centrifugal force, Fe-Ni segregation, and the Taylor-Proudman columns are — but it is the *catalyst* that keeps the window open and sustains the axial oscillation that drives obliquity to its maxima.

7.5 Tidal forces

Solar tide. Relative amplitude $\approx 10^{-4} g_0$; contributes as a low-frequency resonant forcing on obliquity.

Growing proto-lunar tide — Mathieu resonance. From Episode 2 onward, the proto-satellite exerts a tidal force that modulates the stability of the CMT :

$$\ddot{\xi}_r = \left[\lambda + A_m^{(k)}(t)\cos(n_{\text{Moon}}\,t)\right]\xi_r, \tag{17}$$

which is the Mathieu equation (McLachlan, 1947). Parametric resonance bands exist for $n_{\text{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$: the growing proto-lunar tide is the dominant trigger for Episodes 2 to N.

Giant planet tides. Primarily Jupiter : a low-frequency perturbation on obliquity, sustaining the oscillation on timescales of 10^5–10^6 yr.

7.6 Parametric elliptical instability as amplifier of the CMT

In a precessing fluid ellipsoid, precessional flows can resonate with inertial modes of the fluid (Poincaré waves, frequencies $\omega_{\text{in}} = 2\Omega m/n$). This mechanism — the parametric elliptical instability — was established theoretically by Malkus (1968), formalized by Kerswell (2002), and demonstrated experimentally by Lacaze et al. (2004).

For the Maclaurin spheroid geometry at $T_{\text{rot}} = 3.5$ h and $f \approx 0.075$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5}$ rad s^{-1} (Eq. 7). This frequency is commensurable with the inertial modes of the fluid by geometric construction — and not by parameter adjustment. The linear growth rate is :

$$\sigma_{\text{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5}\,\text{rad}\,\text{s}^{-1}. \tag{18}$$

The CMT speed grows exponentially :

$$U_{\text{CMT}}(t) = U_{\text{turb}}\, e^{\sigma_{\text{res}} t}, \qquad U_{\text{turb}} \approx 0.8\,\text{km}\,\text{s}^{-1}, \tag{19}$$

where U_{turb} is the characteristic speed at the Rhines scale (Rhines, 1975). The time to reach $U_{\text{crit}} \approx 9.8$ km s^{-1} is :

$$t_{\text{acc}} = \frac{\ln(U_{\text{crit}}/U_{\text{turb}})}{\sigma_{\text{res}}} \approx 37\,\text{h}, \tag{20}$$

with $\ln(U_{\text{crit}}/U_{\text{turb}}) = \ln(9.8/0.8) \approx 2.51$.

The CMT reaches the bifurcation speed in about 37 hours. This result is entirely determined by $f \approx 0.075$ and $\Omega = 4.99 \times 10^{-4}$ rad s^{-1}, imposed by $T_{\text{rot}} = 3.5$ h and the physics of the Maclaurin spheroid. The elliptical instability thus acts as a powerful amplifier, bringing the CMT to the bifurcation speed in about 37 hours — a timescale fully consistent with the TPT chronology.

As Fe-Ni segregation progresses, the CMT density decreases :

$$\rho_{\mathrm{CMT}}(t) = \rho_0 - \xi_{\mathrm{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4{,}500\,\mathrm{kg\,m^{-3}},\ \rho_f \approx 3{,}200\,\mathrm{kg\,m^{-3}}, \tag{21}$$

which amplifies all destabilizing terms via the factor $\rho_0/\rho_{\mathrm{CMT}}(t)$ in the instability equation (Section 9.2).

7.7 β geostrophic force and Rhines scale

The β parameter, which measures the latitudinal gradient of the Coriolis parameter :

$$\beta = \frac{2\Omega\cos\varepsilon}{R_{\mathrm{eq,proto}}} \approx 7.7\times10^{-11}\,\mathrm{m^{-1}\,s^{-1}} \quad (\varepsilon = 55^\circ,\ R_{\mathrm{eq,proto}} = 7.4\times10^6\,\mathrm{m}), \tag{22}$$

arrests the inverse cascade at the Rhines scale (Rhines, 1975) :

$$L_\beta = \sqrt{\frac{U_{\mathrm{turb}}}{\beta}} \approx 3.0\times10^6\,\mathrm{m} \approx 0.47\,R_\oplus, \tag{23}$$

where U_{turb} is the turbulent velocity scale. This scale $L_\beta \approx 0.47\,R_\oplus$ corresponds to the harmonic mode $\ell = 1, m = 1$: the largest coherent structure possible on a sphere, and the spatial expression of the CMT.

8 Hierarchy of hypotheses

The following table distinguishes established physical consequences (Level 1), well-constrained but not yet numerically validated hypotheses (Level 2), and speculative hypotheses awaiting simulation (Level 3).

TABLE 2 – Explicit hierarchy of hypotheses underlying the TPT.

ID	Hypothesis / Statement	Level	Depends on
H0	Central thesis : the Moon formed by three coupled transitions triggered by Fe-Ni segregation	2 (constrained)	Thermodynamic bound $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 121$

ID	Hypothesis / Statement	Level	Depends on
H1	Total volumetric melting : the entire silicate mantle above the liquidus during the window 4.568–4.4 Ga	1 (established)	Factor $E_{\text{grav}}/E_{\text{fus}} \approx 121$ (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015)
H2	$\varepsilon \in [40°, 70°]$: sustained free nutation in the body's co-rotating frame	2 (constrained)	Absence of Moon + 5 convergent arguments (Laskar et al., 1993a; Touma & Wisdom, 1993; Li & Batygin, 2014)
H3	Confinement of the CMT to $\|\phi\| < 30°$ admits an effective description as the fundamental mode of a nonlinear Schrödinger-type equation with potential $V_{\text{eff}}(\phi)$	3 (speculative)	Reduction from the potential vorticity equation (Redekopp, 1977; Luo, 1991; Yin et al., 2019); conditions $\tau_y \lesssim 10^2$ Pa, $\varepsilon > 45°$; quantitative validity awaiting simulation (Limitation L12)

ID	Hypothesis / Statement	Level	Depends on
H4	The $\ell = 1$ mode is selected by the combination of BH rheology (suppression of modes $\ell < \ell_{\min}$) and inverse cascade	3 (speculative)	3D SPH simulation with full BH rheology (L1)
H5	$\tau_y(T)$ follows an Arrhenius law with $T^* \approx 1,800$ K and $\tau_y \lesssim 10^2$ Pa for $T \gtrsim 4,000$ K	1 (established)	Laboratory rheology (Giordano et al., 2008)
H6	The ejection threshold is defined by crossing $\lambda(U) = 0$; the generalized instability coefficient governs the transition between confinement and ejection	2 (constrained)	Coupled system (Eqs. 26, 28); P22
H7	The effective yield stress τ_y ensures the cohesive integrity of the ejected toroidal flow via the $\mathrm{Bi}_{\mathrm{KH}}$ criterion	2 (constrained)	Frigaard & Nouar (2017); three independent confinement mechanisms
H8	$f_{\mathrm{cap}} \approx 0.70$ and $\partial f_{\mathrm{cap}} / \partial M_{\mathrm{Moon}} > 0$	3 (speculative)	3D SPH simulation (L2); lower bound $f_{\mathrm{cap}}^{\min} \gtrsim 0.45$ (Appendix G)
H9	Dynamo delay = 350 ± 50 Ma, sum of three independent phases	2 (constrained)	Hf-W (Kleine et al., 2002; Yin et al., 2002) + zircons (Wilde et al., 2001; Valley et al., 2002) + Tarduno et al. (2025)

ID	Hypothesis / Statement	Level	Depends on
H10	Seismic preservation window : 200–530 km	2 (constrained)	Ilmenite cumulate overturn (Briaud et al., 2023) + mare volcanism (Li et al., 2021; Che et al., 2021)
H11	Biphasic structure of the CMT : mobile plastic core + fragmented clinker crust	3 (speculative)	3D SPH simulation (L1, L3)
H12	$\mathrm{Bi}_{\mathrm{KH}} \gg 1$ suppresses Kelvin-Helmholtz fragmentation at large and intermediate scales	2 (constrained)	Frigaard & Nouar (2017) ; three independent confinement mechanisms
H13	$b' = 1$ is a dynamical attractor of the coupled segregation–ejection system	2 (constrained)	Geostrophic equilibrium + angular momentum conservation (Appendix B)
H14	The inverse cascade is arrested at the Rhines scale $L_\beta \approx 0.47\, R_\oplus$; the peak at $\ell \approx 14$ is redistributed toward $\ell = 1$ by nonlinearity and BH rheology	3 (speculative)	3D SPH simulation (L1)
H15	The ejected flow transits through three phases : ballistic expansion $\rightarrow$ orbital insertion $\rightarrow$ cohesive accretion beyond R_{Roche}	3 (speculative)	3D SPH simulation (L6)
H16	Parametric elliptical instability amplifies the CMT with growth rate $\sigma_{\mathrm{res}} = \Omega f / 2$	2 (constrained)	Malkus (1968); Kerswell (2002); Lacaze et al. (2004) ; Maclaurin geometry

ID	Hypothesis / Statement	Level	Depends on
H17	The T-Tauri phase maintains $\tau_y \lesssim 10^2$ Pa by irradiation, making the $\ell = 1$ mode possible	2 (constrained)	Stefan-Boltzmann law + BH rheology ; testable by simulation

9 The three coupled transitions

The unique engine is the progressive segregation of Fe-Ni toward the forming core. It simultaneously governs three regime changes : the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the Earth's dynamo.

9.1 Transition 1 — Rheological : the Bingham-Herschel law

A Newtonian fluid flows under any applied stress. A Bingham-Herschel (BH) fluid, by contrast, resists like a solid as long as the applied stress does not exceed a plasticity yield stress τ_y (Bingham, 1922; Herschel & Bulkley, 1926) :

$$\tau = \tau_y + k\dot{\gamma}^n \quad (\tau > \tau_y), \tag{24}$$

where τ is the applied shear stress, τ_y the yield stress, k the consistency index, $\dot{\gamma}$ the shear rate, and n the flow index. For a partially crystallized silicate magma at a solid fraction $\phi \approx 0.1$–0.3 : $n = 1.2$ (shear-thickening, Caricchi et al. 2007), $\tau_y \in [10^2, 10^4]$ Pa at $T = 3,000$–$3,500$ K (Giordano et al., 2008).

The yield stress depends exponentially on temperature (Arrhenius law) :

$$\tau_y(T) = \tau_y^{(0)} \exp\left(+\frac{E_a}{RT}\right), \qquad E_a \approx 150\text{–}250\,\mathrm{kJ\,mol^{-1}}, \tag{25}$$

where R is the gas constant and E_a the activation energy — the positive sign in the exponential reflects that τ_y *increases* as T *decreases*, as for the viscosity of a liquid (Giordano et al., 2008). At $T = 3,000$ K : $\tau_y \approx 10^2$ Pa — the flow moves easily. At $T^* \approx 1,800$ K, depending on the value adopted for E_a within the range 150–250 $\mathrm{kJ\,mol^{-1}}$, τ_y rises to 5.5×10^3–8.0×10^4 Pa — an increase of one to three orders of magnitude compared to $T = 3,000$ K, sufficient for cohesive ejection to become progressively more difficult as the T-Tauri Sun exits its active phase and the surface cools.

This is the natural thermal shutdown mechanism at $T^* \approx 1,800$ K : the TPT window closes irreversibly by the Arrhenius law alone, without additional hypothesis. This is not an adjusted parameter : it is a thermodynamic consequence of the exponential dependence of τ_y on temperature.

The 'a'ā structure of the CMT

The toroidal flow exhibits a biphasic architecture : a mobile plastic core at high temperature surrounded by a fragmented clinker crust. This is not a cohesion problem : it is its guarantee. The external crust protects the core against Kelvin-Helmholtz instabilities at the flow-atmosphere interface. A shell of quenched glass (analogous to the glassy margins of pillow lavas, Hekinian et al. 1989), forming in about 7 minutes upon contact with the atmosphere, completes this rheological shield. This structure is consistent with the work of Frigaard & Nouar (2017) and Balmforth & Craster (2001).

9.2 Transition 2 — Mechanical : dynamical instability and ejection

9.2.1 Principle : a dynamical instability

The ejection of matter from the CMT is a *deterministic dynamical instability* : when the resultant of radial accelerations acting on a fluid parcel becomes repulsive, the parcel accelerates outward and leaves the torus. A parcel undergoing an infinitesimal radial displacement ξ_r obeys the Solberg-Høiland parcel displacement equation (Solberg, 1936; Høiland, 1941) :

$$\ddot{\xi}_r = \lambda\, \xi_r, \qquad \begin{cases} \lambda < 0 & \text{: oscillation, stable confinement,} \\ \lambda > 0 & \text{: exponential growth, ejection.} \end{cases} \tag{26}$$

The ejection threshold is therefore the condition $\lambda = 0$. This threshold is a well-defined surface in parameter space $(\Omega, \varepsilon, \alpha, \beta)$, which gives the mechanism a *predictive* character : for given conditions, ejection occurs or does not occur unambiguously.

9.2.2 The stability coefficient λ

In solid-body rotation ($U = \Omega r$), with $U \propto r^{\alpha}$ the azimuthal velocity profile, the stability coefficient gathers all radial accelerations acting on the parcel :

The instability coefficient

$$\lambda = \underbrace{-\frac{g_{\text{eff}}}{r} - N^2}_{\text{constant terms}} + \Omega^2 \underbrace{\left(\overbrace{2\sin\varepsilon_{\text{dyn}}}^{\text{Eötvös}} - \overbrace{2(1+\alpha)}^{\text{Rayleigh}} + \overbrace{\beta}^{\text{centrifug. buoyancy}} \right)}_{D} \tag{27}$$

Each term has a precise physical origin — and a simple history :

— $-g_{\text{eff}}/r$: confining effective gravity, always *stabilizing*; this is the term to overcome.

— $-N^2$: *thermal* buoyancy (Brunt–Väisälä frequency squared, Väisälä, 1925; Brunt, 1927). The magma is cooled from above : the thermal gradient makes the medium superadiabatic, so $N^2 < 0$ and $-N^2 > 0$; this thermal part is *destabilizing* and sustains convection. The *compositional* part of the density gradient is carried separately by β (Ledoux-type separation, Appendix C).

— $2\sin\varepsilon_{\text{dyn}}$: *non-traditional* component of the Coriolis force (Eötvös effect) : any prograde flow is radially lightened by $2\Omega U\cos\phi$, where ϕ is the latitude. Here ε_{dyn} is the *colatitude of the CMT relative to the instantaneous dynamical equator* ($\sin\varepsilon_{\text{dyn}} = \cos\phi$) — no orbital reference enters the internal dynamics. Neglected in the "traditional approximation" of geophysics, this component must be retained at low latitudes (Gerkema et al., 2008) : this is precisely the regime of the intertropical belt. Term *destabilizing*, maximal when the torus straddles the dynamical equator ($\varepsilon_{\text{dyn}} \to 90°$).

— $-2(1+\alpha)$: Rayleigh term (Rayleigh, 1917); the epicyclic frequency $\kappa^2 = r^{-3}\,\mathrm{d}[(rU)^2]/\mathrm{d}r = 2(1+\alpha)U^2/r^2$ is always *stabilizing*, hence the negative sign. This centrifugal charge gradient contains *by itself* the entire centrifugal effect : no separate "centrifugal pressure" term should be added to it, on pain of double counting (Appendix C).

— $+\beta$: *centrifugal buoyancy* on the density gradient, $\beta \equiv -\mathrm{d}\ln\rho/\mathrm{d}\ln r$. In the CMT, $\beta > 0$: Fe-Ni segregation makes the body denser at depth. With respect to gravity, this stratification is stable; but the centrifugal force is a gravity that points *outward*, and for it, the same stratification is a Rayleigh-Taylor-type configuration : the denser fluid is pushed outward. Term *destabilizing* — and essential : it is the engine's fuel. Fe-Ni segregation thus does double duty : it accelerates rotation by contraction (ice-skater effect) and it creates the stratification whose centrifugal buoyancy fuels the ejection.

9.2.3 *The rotation threshold*

Setting $\lambda = 0$, one isolates the critical angular speed :

$$\Omega_{\text{crit}}^2 = \frac{g_{\text{eff}}/r + N^2}{2\sin\varepsilon_{\text{dyn}} - 2(1+\alpha) + \beta}. \tag{28}$$

A finite threshold exists only if the denominator D is positive, i.e., if $\beta > 2(1+\alpha) - 2\sin\varepsilon_{\text{dyn}}$. The driving role of β is directly read here : with the nominal values, $2\sin\varepsilon_{\text{dyn}} - 2(1+\alpha) = 1.879 - 3 = -1.12$ — without centrifugal buoyancy, no rotation would ever trigger ejection. With $\beta = 2.5$ ($\varepsilon_{\text{dyn}} = 70°$ at the nutation peak, $\alpha = 0.5$, $N^2 = -1.93 \times 10^{-7}\ \text{s}^{-2}$), the denominator is $D = 1.379$ and the threshold $\Omega_{\text{crit}} \approx 1.27\,\Omega_0$, reachable by angular momentum input without approaching the critical rotation regime.

9.2.4 *The episode cycle*

The mechanism is a **deterministic relaxation cycle** :

1. Ω grows slowly by net angular momentum input (ice-skater effect of Fe-Ni segregation, complemented by continuous accretion inputs).
2. The axis nutates (period $\approx$ 46.7 h) : the effective colatitude $\varepsilon_{\mathrm{dyn}}$ oscillates in $[40°, 70°]$. At the nutation peak, the threshold Ω_{crit} is minimal.
3. A major Late Veneer impact (body $\gtrsim$ 750 km) occurs during the post-ejection plateau window (Appendix I). The impulsive torque of the impact provides a jump $\Delta\Omega$, and ε is already close to its maximum. The combination of the two lowers Ω_{crit} and crosses the threshold.
4. A magmatic tongue stretches, pinches off by Rayleigh–Plateau instability (Plateau, 1873; Rayleigh, 1878), delayed by Bingham cohesion, and detaches tangentially : an ejection episode.
5. The ejection carries away a fraction η of the angular momentum ; Ω falls back below the threshold, then the system recharges until the next major impact.

9.2.5 *Role of cohesion : the* $\mathrm{Bi}_{\mathrm{KH}}$ *criterion*

The Bingham yield stress τ_y **does not intervene** in the ejection threshold (Eq. 28) : at the torus scale, it is negligible compared to gravity. Its role is distinct : it maintains the cohesion of the ejected flow, which propagates as a *continuous tongue*, via the Bingham–Kelvin–Helmholtz criterion $\mathrm{Bi}_{\mathrm{KH}} \gg 1$ (Appendix D). Cohesion of the flow and ejection threshold are thus physically separate questions.

9.2.6 *Critical speed and maximum ejectable mass*

The critical ejection speed, derived from the energetic escape condition, is :

$$U_{\mathrm{crit}}^{(0)} = \sqrt{2\, g_{\mathrm{eff}}^{(0)}\, R_{\mathrm{eq,proto}}} \approx 9.8\,\mathrm{km\,s^{-1}}, \tag{29}$$

which is consistent with the escape energy scale. The Mach number at ejection, using the sound speed in the biphasic magma $c_s \approx 0.9\ \mathrm{km\,s^{-1}}$ (Mader et al., 2013), is $\mathrm{Ma} = U_{\mathrm{crit}}/c_s \approx 10.9$: the ejection is **hypersonic**.

The maximum ejectable mass per episode is the fraction of the CMT with $U > U_{\mathrm{crit}}$:

$$M_{\mathrm{ej}}^{\max} \approx f_{\mathrm{inst}}\, M_{\mathrm{CMT}}, \tag{30}$$

where f_{inst} is the unstable fraction of the velocity distribution. For a typical rotation profile, this fraction is approximately 0.1, giving :

$$M_{\mathrm{ej}}^{\max} \approx 2.0\text{–}4.0 \times 10^{22}\,\mathrm{kg}, \tag{31}$$

consistent with the mass budget (Section 10). This is why $N = 2$–3 episodes suffice.

9.2.7 *Cohesion of the ejected flow : the* $\mathrm{Bi}_{\mathrm{KH}}$ *criterion*

For a free flow, the Weber number $We \approx 1.4 \times 10^{15}$ would imply fragmentation by atomization. However, the CMT is not a free flow : it is a collective, self-gravitating toroidal flow, confined by toroidal symmetry, self-gravity, and atmospheric pressure.

The correct criterion is the Bingham-Kelvin-Helmholtz number, which compares the yield stress to the Kelvin-Helmholtz shear stress : for $\mathrm{Bi}_{\mathrm{KH}} \gg 1$, the yield stress suppresses the KH instability and cohesion is maintained (Frigaard & Nouar, 2017). Three independent conditions reinforce cohesion : (1) atmospheric confinement ($P \approx 10^5$–10^7 Pa) ; (2) internal crystal network ($\phi \approx 0.1$–0.3, multiplying fragmentation resistance by a factor of 3 to 10) ; (3) density ratio $\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \approx 200$ (Rayleigh-Taylor suppression). The complete demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ criterion is given in Appendix D.

Effective cohesion reinforcements

The effective cohesion is the product of the three reinforcements :

$$\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} = \mathrm{Bi}_{\mathrm{KH}} \times f_{\mathrm{atm}} \times f_{\mathrm{cryst}} \times f_{\mathrm{RT}} \tag{32}$$

with :

- $f_{\mathrm{atm}} \sim 10$–100 (atmospheric confinement) - $f_{\mathrm{cryst}} \sim 3$–$10$ (internal crystal network) - $f_{\mathrm{RT}} \sim 10$–50 (Rayleigh-Taylor suppression)

With the nominal values, $\mathrm{Bi}_{\mathrm{KH}} \sim 3 \times 10^{-2}$ and the product of the three reinforcements is ~ 300, giving $\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} \sim 9$, much greater than 1. Cohesion is therefore assured.

Recognized limitation : this calculation is a nominal estimate. A 3D SPH simulation with full BH rheology is necessary to validate the exact values, in particular the nonlinear interaction between the three effects (Limitation L14, Section 18).

9.2.8 *Post-ejection transient precession and crustal dichotomy*

The ejection of $M_{\mathrm{ej}} \approx 2$–$4 \times 10^{22}$ kg from the intertropical belt is not dynamically negligible. The asymmetric mass loss modifies the inertia tensor, producing : (1) a decrease $\Delta\Omega/\Omega \approx -10$–$15\%$, lowering g_{eff} and U_{crit} for the following episode ; (2) shape oscillations as the spheroid contracts toward a rounder sphere ; (3) a reorganization of obliquity toward a new value within $[40°, 70°]$.

This last point has a direct geometric consequence, and not merely a numerical one : since the intertropical belt $|\phi| < 30°$ is defined perpendicular to the instantaneous axis $\boldsymbol{\Omega}(t)$ (Section 3), a new value of ε displaces the position of the dynamical equator $\phi = 0$ itself, and therefore the entire active belt, relative to the body. The intertropical belt hosting the second episode is therefore not, in general, the same region of the body as that hosting the first : same angular width ($\pm 30^\circ$), same relative definition with respect to $\mathbf{\Omega}(t)$, but a different location on the spheroid. It is this geometric displacement of the active belt, episode after episode, that constitutes the direct cause of the crustal asymmetry described below, and not merely the change in the numerical value of ε.

This displacement of the active belt, combined with the shape oscillations from point (2) above, has an additional consequence for the CMT itself : the torus that reforms for the following episode does not reproduce an identical copy of the previous one simply displaced. It forms on a spheroid whose own shape has slightly changed, along a belt whose position has itself changed. The subsequent CMT is therefore ejected toroidally in a manner distinct from the previous one, both in shape and orientation. This remains, at this stage, a qualitative statement : no quantitative derivation of the shape or angular shift between successive episodes is attempted here.

The second episode ejects a cohesive toroidal flow onto a still hot and deformable proto-POB, within a limited solid angle (corresponding to the projected intertropical belt), which becomes the present farside. Three mechanisms amplify the resulting crustal asymmetry : early gravitational locking ($\tau_{\rm lock} \sim 10^4$–$10^5$ yr (Wieczorek et al., 2013)), residual obliquity of the proto-POB, and irreversible BH plastic welding. The thickness ratio $\approx 2{:}1$ (nearside $\approx$ 34 km vs farside $\approx$ 54 km, Wieczorek et al. 2013) is a qualitative prediction of the TPT.

9.3 Transition 3 — Magnetic : the progressive dynamo

The delay of about 350 Ma between the inferred time of lunar formation ($\approx$ 4.55 Ga) and the first geological detection of Earth's magnetic field ($\approx$ 4.2 Ga, Tarduno et al. 2025) is not an adjusted parameter in the TPT. It is proposed as the sum of three independently constrained durations :

$$\Delta t_{\rm dynamo} = \underbrace{\Delta t_{\rm ej}}_{\approx 60\,\rm Ma} + \underbrace{\Delta t_{\rm refr}}_{\approx 140\,\rm Ma} + \underbrace{\Delta t_{\rm em}}_{\approx 150\,\rm Ma} \approx 350\,\rm Ma. \tag{33}$$

Phase 1 ($\approx$ 60 Ma) — Active Fe-Ni segregation. Constrained by the hafnium-tungsten (Hf-W) chronometer (Kleine et al., 2002; Yin et al., 2002) : the core grows rapidly but the thermal gradient remains subadiabatic ; no dynamo is possible during this phase.

Phase 2 ($\approx$ 140 Ma) — Protocrust formation. The growing Moon stabilizes obliquity to $\approx 23.5^\circ$; the protocrust forms around 4.4 Ga, as attested by Jack Hills zircons (Wilde et al., 2001; Valley et al., 2002).

Phase 3 ($\approx$ 150 Ma) — Progressive emergence of the dynamo. Compositional convection in the Fe-Ni core grows until the Lorentz force becomes comparable to the

Coriolis force. The Elsasser number Λ balances these two forces. The threshold $B_{\text{crit}} \approx 2.6$ mT is derived in Appendix E. Modern geodynamo simulations (Christensen & Aubert, 2006; Olson & Christensen, 2006) show that the transition is progressive, not abrupt : $\Lambda \sim 1$ is an order-of-magnitude reference, not a strict threshold.

Section synthesis

The delay of about 350 Ma involves no free parameter : Δt_{ej} is constrained by the Hf-W chronometer (Kleine et al., 2002; Yin et al., 2002) ; Δt_{refr} is constrained by Jack Hills zircons at 4.4 Ga (Wilde et al., 2001; Valley et al., 2002) ; and Δt_{em} is constrained by Tarduno et al. (2025) at 4.2 Ga. These are three independent observations, not an interpolation. The associated uncertainty is ± 50 Ma (Limitation L4, Section 18).

10 Mass budget : $N = 2$–3 episodes

10.1 Corrected target lunar mass

A fundamental point : the presently observed lunar mass $M_{\text{Moon}} = 7.34 \times 10^{22}$ kg is *not* the mass that the TPT mechanism must produce. The Moon accreted additional mass after the ejection episodes : a differentiated body about 260 km in diameter struck the Moon at the South Pole–Aitken basin and *remained there*, contributing $M_{\text{SPA}} \approx 3.2 \times 10^{19}$ kg (Wakita et al., 2026) — this material is now part of the observed lunar mass — and documented late accretion contributed an additional mass ΔM_{late} (Zellner, 2019). The target mass for the TPT mechanism is therefore :

$$M_{\text{TPT}} = M_{\text{Moon}} - M_{\text{SPA}} - \Delta M_{\text{late}} < 7.34 \times 10^{22}\,\text{kg}. \tag{34}$$

10.2 Mass budget equation

The mass budget equation

$$M_{\text{Moon}} = N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\text{max}} + M_{\text{SPA}} + \Delta M_{\text{late}}, \tag{35}$$

with $N = 2$–3 (nominal), $f_{\text{cap}} \in [0.65, 0.85]$ (nominal 0.70), $M_{\text{ej}}^{\text{max}} \approx 2.0$–$4.0 \times 10^{22}$ kg per episode.

f_{cap}	$M_{\text{TPT}}^{(N=2)}$	$M_{\text{TPT}}^{(N=3)}$	Fraction of M_{Moon}
0.50	2.5×10^{22} kg	3.7×10^{22} kg	34–50%
0.70	3.5×10^{22} kg	5.2×10^{22} kg	48–71%
0.85	4.3×10^{22} kg	6.4×10^{22} kg	59–87%

The remainder is accounted for by M_{SPA} and documented late accretion. The budget is robust over the entire f_{cap} range.

f_{cap} is an increasing state variable : $\partial f_{\text{cap}}/\partial M_{\text{Moon}} > 0$, because the larger the proto-Moon, the more efficiently it captures subsequent ejecta. This positive feedback, combined with the decrease in $g_{\text{eff}}^{(k)}$ at each episode, defines a dynamical attractor that reinforces the mechanism.

10.3 f_{cap} as a testable prediction, not a free parameter

The position adopted here is that f_{cap} should not be treated as an adjustable parameter whose broad range would make the mass budget uninformative, but as a quantity that the TPT predicts quantitatively and that can, in principle, falsify the mechanism.

Physical basis : the classical disk accretion range. The reference N-body simulations of lunar accretion from an impact- generated debris disk (Kokubo et al., 2000) establish that the efficiency of incorporation of disk matter into a final Moon is 10 to 55%, this efficiency increasing with the initial specific angular momentum of the disk. This result, robust over a wide range of initial conditions and numerical resolutions, constitutes the external reference to which f_{cap} must be compared.

The deviation as a potential signature. The nominal value $f_{\text{cap}} \approx 0.70$ adopted in the TPT is significantly higher than the upper bound of this classical range (55%). Within the TPT framework, this deviation is not accidental : it is attributed to the coherence of the ejected toroidal flow and its orbital concentration within a limited solid angle (Appendix G), as opposed to a debris disk dispersed over 4π steradians as in the classical giant-impact scenario. If this interpretation is correct, the deviation between the f_{cap} value adopted by the TPT and the classical range becomes a discriminating test rather than a mere parametric latitude.

Falsifiable prediction — P17 — Capture efficiency beyond the classical disk accretion range

The orbital coherence of the toroidal flow ejected by the TPT pushes the effective capture efficiency beyond the classical range of 10–55% established for accretion from a dispersed debris disk (Kokubo et al., 2000). The TPT predicts $f_{\text{cap}} \gtrsim 0.65$–$0.70$.

Falsification : if a value of f_{cap} derived from a dedicated 3D SPH simulation of the CMT geometry (BH rheology, atmospheric confinement, orbital concentration within the intertropical belt) falls within the classical range of 10–55% rather than above it, this aspect of the theory is falsified. A f_{cap} measured in the classical range would not necessarily invalidate the TPT as a whole (the mass budget would remain achievable via M_{SPA} and late accretion), but it would deprive f_{cap} of its status as a distinctive signature relative to the giant-impact model.

Recognized limitation : this prediction rests on the assumption that the orbital concentration of the TPT flow (Appendix G) produces a qualitatively different accretion regime from the classical dispersed disk. This assumption is itself Level 3 (not quantitatively demonstrated), so this does not become a robust test until the 3D SPH simulation is available. In the meantime, this remains a qualified prediction : the deviation from the classical range is consistent with the TPT, but does not independently confirm it.

11 Angular momentum transport by ejecta

In the present framework, angular momentum loss is not treated as a single-speed ballistic process, but as a distributed transport mechanism resulting from a spectrum of ejected material generated over successive phases driven by differentiation.

11.1 Ejecta distribution function

We describe the mass distribution of ejecta as a function of velocity and emission angle :

$$dM = \rho(v, \theta)\, dv\, d\theta, \tag{36}$$

where v is the ejection velocity, θ is the angle relative to the local tangential direction, and $\rho(v, \theta)$ is the ejecta distribution function depending on the phase.

The distribution $\rho(v, \theta)$ is not arbitrary. It follows from three physical processes : the CMT velocity spectrum is not monokinetic, with turbulent fluctuations at the Rhines scale (Section 7.6) producing a velocity spread $\delta v_{\mathrm{CMT}} \sim U_{\mathrm{turb}} \approx 0.8\ \mathrm{km\,s^{-1}}$; and the finite extension of the torus produces an angular spread $\rho_\theta(\theta) \propto \cos\theta$ for $|\theta| < \theta_{\max}$, with $\theta_{\max} \approx 30°$.

For analytical tractability, we adopt the separable form :

$$\rho(v, \theta) = M_{\mathrm{ej}}\, \mathcal{N}_v\, \mathcal{N}_\theta\, \exp\left[-\frac{(v - \bar{v})^2}{2\sigma_v^2}\right] \cos\theta, \quad v > v_{\mathrm{esc}},\ |\theta| < \theta_{\max}, \tag{37}$$

where $v_{\mathrm{esc}} = \sqrt{2 g_{\mathrm{eff}} R_p}$. No free parameter is introduced beyond those already present in the theory developed in the preceding sections.

11.2 Specific angular momentum

The specific angular momentum of an ejecta element is $h = r\, v \sin\theta$, where r is the effective emission radius. This radius is not simply the planetary radius R_p : the silicate atmosphere (Section 2) extends the effective surface to $r_{\mathrm{atm}} \approx R_p + H$, with scale height $H \approx 200$ km for $T \approx 3000$ K and $g \approx 6\ \mathrm{m/s^2}$; and the toroidal flow is elevated above the surface by centrifugal effects, with the CMT center at $r_{\mathrm{CMT}} \approx R_p + U_{\mathrm{CMT}}^2/(2 g_{\mathrm{eff}}) \approx R_p + 500$ km. We therefore adopt $r \approx R_p + \Delta r$, with $\Delta r \approx 300$–500 km. This correction introduces only a few percent uncertainty in the angular momentum budget, well within the overall error margin.

11.2.1 Assembly beyond the Roche radius

An apparent paradox deserves to be resolved here, for it conditions the very birth of the Moon. The maximum specific angular momentum that an ejecta can carry at launch, $h_{\max} \approx 7.74 \times 10^{10}$ m^2 s^{-1}, corresponds to a circular orbit of radius $r_{\rm circ} = h^2_{\max}/(GM_p) \approx 1.5 \times 10^7$ m, i.e., about 15,000 km — *below* the Roche radius ($\approx 22,200$ km), where tidal forces prohibit direct accretion of a self-gravitating body (Roche, 1849). Taken in isolation, no captured ejecta can therefore become a moon : the matter seems condemned to orbit in a ring, torn apart by tides.

The resolution of this paradox is known, and it requires no new hypothesis : the captured torus is not a set of particles frozen on independent Keplerian orbits, but a dense disk where inelastic collisions and gravitational interactions generate an effective viscosity. This viscosity does what any disk viscosity does : it transports angular momentum outward and mass in both directions, so that a fraction of the disk is pushed beyond the Roche radius — where accretion becomes permitted again and where the proto-Moon assembles, fed by the disk edge. This spreading mechanism is precisely the one that builds the Moon in post-impact disk models : in the particulate regime, assembly is completed in a few months to a year (Ida et al., 1997) ; in the fluid disk regime, where radiative cooling regulates viscosity, it extends over $\sim 10^2$–10^3 yr (Salmon & Canup, 2012). In both cases, assembly is completed well before the evection resonance encounter (10^3–10^5 yr, Section 11.5) : the Moon exists as a single body when the angular momentum sink engages.

This result has an important conceptual consequence for the TPT : the cohesion of the ejected flow ($\mathrm{Bi_{KH}}$ number, Appendix D, Limitation L14) governs the *launch and transport* phase — how much matter reaches orbit, and with what angular momentum — but it is not required for the *assembly* phase, which viscous spreading accomplishes alone. The critical question posed to the SPH simulation is therefore not "does the flow remain perfectly cohesive to orbit ?" but "what fraction of the ejected mass is captured in stable orbit ?" — i.e., the determination of $f_{\rm cap}$ (Limitation L2).

11.2.2 Energy budget of ejection

With the angular momentum budget now in place, the symmetric question naturally arises : does the proto-Earth have the *energy* to launch three magmatic tongues at escape speed ? The total kinetic energy of the ejecta is :

$$E_{\rm kin,ej} = \frac{1}{2}\, N M_{\rm ej}\, \bar{v}^2 \approx \frac{1}{2} \times 3 \times 2.5 \times 10^{22} \times (9.8 \times 10^3)^2 \approx 3.6 \times 10^{30}\,\mathrm{J}, \qquad (38)$$

most of which is spent climbing the potential well ($\bar{v}$ being close to escape speed, the residual kinetic energy at infinity is small). The available reservoir is the rotational energy of the proto-Earth :

$$E_{\rm rot} = \frac{1}{2}\, I_p\, \Omega_p^2 \approx \frac{1}{2} \times 1.32 \times 10^{38} \times (4.99 \times 10^{-4})^2 \approx 1.6 \times 10^{31}\,\text{J}. \tag{39}$$

The ratio $E_{\rm kin,ej}/E_{\rm rot} \approx 0.22$: the three episodes consume about one fifth of the spin reservoir.

The total angular momentum carried away *at launch* by all ejecta, whether subsequently captured or escaping, is :

$$L_{\rm ej} = M_{\rm ej}\, r\, \bar{v}\, \langle \sin\theta \rangle \approx 7.5 \times 10^{22} \times 7.9 \times 10^{6} \times 9.8 \times 10^{3} \times 0.25 \approx 1.5 \times 10^{33}\,\text{J}\cdot\text{s}, \tag{40}$$

where $\langle \sin\theta \rangle \approx 0.25$ is the mean geometric factor accounting for the angular spread of the torus ($\theta_{\rm max} \approx 30°$). This quantity $L_{\rm ej}$ is distinct from $L_{\rm loss}$ as formally defined in Eq. 43 (Section 11.3) : $L_{\rm ej}$ covers all ejecta at launch, while $L_{\rm loss}$ covers only the uncaptured fraction. Weighting by the escaping fraction $(1 - f_{\rm cap}) = 0.30$ gives a mean estimate of $L_{\rm loss}$ consistent with that formal definition :

$$L_{\rm loss} \approx (1 - f_{\rm cap})\, L_{\rm ej} \approx 0.30 \times 1.5 \times 10^{33} \approx 4.5 \times 10^{32}\,\text{J}\cdot\text{s}. \tag{41}$$

This value is numerically below the conservative bound $L_{\rm esc}^{\rm max} \approx 1.74 \times 10^{33}$ J s (Section 11.5), which instead assumes purely tangential ejection ($\sin\theta = 1$) for the escaping material alone — a deliberately generous upper bound, not a central value. The escape channel therefore covers an even smaller fraction of the deficit ΔL than the conservative bound alone suggests.

The rotational energy loss associated with $L_{\rm ej}$ is :

$$\Delta E_{\rm rot} \approx 2\, \frac{L_{\rm ej}}{L_{\rm init}}\, E_{\rm rot} \approx 2 \times \frac{1.5 \times 10^{33}}{6.59 \times 10^{34}} \times 1.6 \times 10^{31} \approx 7.3 \times 10^{29}\,\text{J}, \tag{42}$$

to which is added, and by which it is dominated, the Fe-Ni segregation heat ($\approx 3 \times 10^{30}$ J, Section 9.2), which constitutes the main source covering the energy bill of ejection and further sustains the molten state and convection. The energy budget is therefore covered to order unity by reservoirs already inventoried in the model : the ejection is energetically feasible without hidden sources, and without compromising the rotational stability of the host body.

11.3 Angular momentum loss

The total angular momentum carried away by non-accreted matter is :

$$L_{\rm loss} = \int\!\!\int \big(1 - f_{\rm cap}(v,\theta)\big)\; r\, v \sin\theta\, \rho(v,\theta)\, dv\, d\theta. \tag{43}$$

The capture efficiency $f_{\mathrm{cap}}(v,\theta)$ is not a constant : it depends on the orbital fate of each ejecta element,

$$f_{\mathrm{cap}}(v,\theta) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{v\sin\theta - v_{\mathrm{crit}}}{\sigma_f}\right)\right], \tag{44}$$

where $v_{\mathrm{crit}} \approx v_{\mathrm{circ}} = \sqrt{g_{\mathrm{eff}} R_p}$ is the circular orbital speed and σ_f a smoothing width representing the finite transition between captured and lost matter. Rather than treating f_{cap} as a single ad hoc value, we express it here as a function $f_{\mathrm{cap}}(v,\theta)$ of orbital mechanics. The nominal value $f_{\mathrm{cap}} \approx 0.70$, estimated independently from the cohesive fraction of the CMT and the re-accretion of co-orbital fragments, is consistent with the mass- weighted average of this function over the proposed ejecta spectrum.

11.4 Discrete formulation and Monte Carlo implementation

For numerical implementation, the expression can be discretized as :

$$L_{\mathrm{loss}} = \sum_i m_i\, r_i\, v_i \sin\theta_i\, \left(1 - f_{\mathrm{cap},i}\right). \tag{45}$$

A stochastic treatment can implement this discrete formulation by Monte Carlo sampling of $\rho(v,\theta)$: sample N particles from $\rho(v,\theta)$, assign each particle a mass $m_i = M_{\mathrm{ej}}/N$, compute $h_i = r_i v_i \sin\theta_i$, compute $f_{\mathrm{cap},i}$ from Eq. 44, and sum $L_{\mathrm{loss}} = \sum_i m_i h_i (1 - f_{\mathrm{cap},i})$. This Monte Carlo approach is recommended for 3D SPH validation (Limitation L2) : it provides a robust check of the analytical estimates.

In a multi-phase formation scenario, the ejecta are further grouped into k distinct formation phases :

$$L_{\mathrm{loss}} = \sum_k \sum_i m_{k,i}\, r_{k,i}\, v_{k,i} \sin\theta_{k,i}\, \left(1 - f_{\mathrm{cap},k,i}\right), \tag{46}$$

each phase k having its own mean ejection velocity $\bar{v}_k$ (decreasing as the proto-Earth loses mass and angular momentum), its own angular spread $\theta_{\max,k}$ (which may increase as the CMT reorganizes between episodes), and its own mean capture efficiency $f_{\mathrm{cap},k}$ (which increases as the proto-Moon grows, the positive feedback of Section 10.3).

11.5 Closure of the angular momentum budget

The total angular momentum budget reads :

$$L_{\mathrm{init}} = L_{\mathrm{TL}} + L_{\mathrm{esc}} + L_{\mathrm{tidal}} + L_{\mathrm{autres}}, \tag{47}$$

where :

— $L_{\text{init}} = 6.59 \times 10^{34}$ J s is the initial angular momentum of the proto-Earth at $T = 3.5$ h ;
— $L_{\text{TL}} = 3.47 \times 10^{34}$ J s is the present angular momentum of the Earth–Moon system (Earth's spin + lunar orbit) ;
— L_{esc} is the angular momentum carried away by ejecta that permanently escape the system ;
— L_{tidal} is the angular momentum evacuated by tidal recession and the evection resonance ;
— L_{autres} regroups losses through other channels (magnetic torque, solar wind, interaction with the protoplanetary disk).

Conservative bound on L_{esc}. For ejecta launched from $r \approx 7.9 \times 10^6$ m at $v \approx 9.8$ km/s (pre-shock speed), the maximum specific angular momentum is $h_{\max} = r\bar{v} = 7.74 \times 10^{10}$ m²/s. With total ejected mass $M_{\text{ej}} = 7.5 \times 10^{22}$ kg and capture efficiency $f_{\text{cap}} = 0.70$, the escaped mass is $M_{\text{esc}} = 2.25 \times 10^{22}$ kg. The upper bound on L_{esc} is therefore :

$$L_{\text{esc}}^{\max} = M_{\text{esc}} h_{\max} \approx 1.74 \times 10^{33}\ \text{J} \cdot \text{s}. \tag{48}$$

This bound is computed with the pre-shock speed. Since the ejection is hypersonic (Ma ≈ 11, Section 9.2), a detached shock forms at the CMT interface, reducing the tangential speed of the ejecta. The actual value of L_{esc} is therefore likely below this bound. It should be understood as a conservative estimate — an upper bound, not a central value.

Deficit and role of the evection resonance. The deficit relative to the present state is :

$$\Delta L = L_{\text{init}} - L_{\text{TL}} - L_{\text{esc}}^{\max} \approx 6.59 \times 10^{34} - 3.47 \times 10^{34} - 1.74 \times 10^{33} \approx 2.95 \times 10^{34}\ \text{J} \cdot \text{s}, \tag{49}$$

i.e., about 44.8% of the initial angular momentum.

With $L_{\text{esc}} \leq 1.74 \times 10^{33}$ J s, the escape channel covers only a minor fraction of the deficit. Most of ΔL must therefore be evacuated by an external sink.

The evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) is the most studied quantified candidate in the literature. It transfers excess angular momentum from the Earth–Moon system to the heliocentric orbit during tidal recession. Its efficiency depends on the quality factor Q of the proto-Earth, hence on its thermal state and solid fraction ϕ. For $Q \sim 10^4$ — a value associated with the "near-totally molten" state ($\phi \sim 0.1$–0.3, Henning et al. 2009) — the resonance crossing time ($\sim 3 \times 10^3$ yr) is shorter than the time required to evacuate ΔL ($\sim 5 \times 10^4$ yr). Only $Q \gtrsim 10^5$–10^6

(near-pure fluid, $\phi \ll 0.01$) offers sufficient margin. Quantitative confirmation of this mechanism, and identification of any additional channels, requires a dedicated simulation of the evolution through the resonance following Rufu & Canup (2020).

Status of the budget. In the current state of the manuscript, the angular momentum budget is not closed for the nominal value of Q associated with the "near-totally molten" state ($Q \sim 10^4$). It is closed for $Q \gtrsim 10^5$, but this value is in tension with the BH rheology used for CMT cohesion (Section 9.1). This tension is identified as the major limitation of the TPT in its present version (Limitation L15, Section 18).

The minimum period compatible with the carry-away bound, given for reference, is :

$$T_{\text{init}} \gtrsim \frac{2\pi I_p}{L_{\text{TL}} + L_{\text{esc}}^{\text{max}}} \approx \frac{2\pi(1.32 \times 10^{38})}{3.47 \times 10^{34} + 1.74 \times 10^{33}} \approx 6.33 \text{ h}. \tag{50}$$

This path is excluded by the mechanism itself : the ejection threshold Ω_{crit} (Section 9.2) is an absolute quantity, and a proto-Earth at $T = 6.33$ h could not reach it.

11.6 Scaling laws and dimensionless form

The angular momentum transport problem admits a natural non-dimensionalization that reveals the essential physical dependences and simplifies the parameter space. We define the characteristic velocity scale $U_0 = U_{\text{crit}} \approx 9.8 \text{ km s}^{-1}$, length scale $R_0 = R_p \approx 7.4 \times 10^6$ m, mass scale $M_0 = M_{\text{ej}} \approx 2.5 \times 10^{22}$ kg, and angular momentum scale $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36} \text{ kg m}^2 \text{ s}^{-1}$. The dimensionless variables $\tilde{v} = v/U_0$, $\tilde{r} = r/R_0$, $\tilde{m} = m/M_0$, and $\tilde{h} = h/(R_0 U_0) = \tilde{r}\tilde{v}\sin\theta$ reformulate the ejecta distribution as :

$$\tilde{\rho}(\tilde{v},\theta) = \tilde{M}_{\text{ej}} \tilde{\mathcal{N}}_v \tilde{\mathcal{N}}_\theta \exp\left[-\frac{(\tilde{v}-\bar{\tilde{v}})^2}{2\tilde{\sigma}_v^2}\right] \cos\theta, \quad \tilde{v} > \tilde{v}_{\text{esc}},\ |\theta| < \theta_{\text{max}}, \tag{51}$$

with $\tilde{M}_{\text{ej}} = 1$ by definition, $\bar{\tilde{v}} = \bar{v}/U_0 \approx 1$, $\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05$–$0.10$, and $\tilde{v}_{\text{esc}} = v_{\text{esc}}/U_0 \approx 1/\sqrt{2}$. The dimensionless capture efficiency takes the same erf form as Eq. 44, with $\tilde{v}_{\text{crit}} \approx 1/\sqrt{2}$, and the dimensionless angular momentum lost, $\tilde{L_{\text{loss}}}$, recovers the physical quantity via $L_{\text{loss}} = H_0\, \tilde{L_{\text{loss}}}$.

This dimensionless form reveals that the problem depends only on four dimensionless parameters : the mean ejection velocity relative to U_{crit} ($\bar{\tilde{v}} \approx 1$), the velocity dispersion ($\tilde{\sigma}_v \approx 0.05$–$0.10$), the angular extension of the torus ($\theta_{\text{max}} \approx 30°$), and the capture threshold ($\tilde{v}_{\text{crit}} \approx 1/\sqrt{2}$). This low-dimensional parameter space keeps the model testable : each parameter can in principle be independently constrained by future simulations or observations.

The dimensionless form admits three asymptotic regimes. In the monokinetic limit

($\tilde{\sigma}_v \to 0$), the distribution reduces to a single speed and recovers the single-speed ballistic approximation used in previous sections. In the broad-spectrum limit ($\tilde{\sigma}_v \gg 1$), the velocity distribution becomes uniform over the available range, maximizing the angular momentum carried by the fastest ejecta. In the sharp-cutoff limit ($\tilde{\sigma}_f \to 0$), the capture efficiency becomes a step function, representing the idealized case where all matter above the capture threshold is retained.

Falsifiable prediction — P20 — Asymptotic regime indicator

The actual ejecta spectrum of the TPT lies between the monokinetic and broad-spectrum limits, with $\tilde{\sigma}_v \approx 0.05$–$0.10$. The sharp-cutoff limit provides an upper bound on L_{loss}, while the monokinetic limit provides a lower bound. The actual value, accessible by Monte Carlo simulation, should lie between these bounds.

Test : Monte Carlo sampling of $\rho(v, \theta)$ (Section 11.4) compared to the analytical monokinetic and broad-spectrum limits derived above.

The dimensionless form also allows derivation of scaling laws for L_{loss} as a function of the fundamental proto-Earth parameters : $L_{\text{loss}} \propto M_p R_p^2 \Omega_p$ (the standard scaling law for the angular momentum of a rotating body), $L_{\text{loss}} \propto \sqrt{\sin \varepsilon}$ (from the obliquity dependence of the ejection speed threshold), and $\delta L_{\text{loss}}/L_{\text{loss}} \approx -\delta f_{\text{cap}}/(1 - f_{\text{cap}})$. For $f_{\text{cap}} = 0.70$, $1 - f_{\text{cap}} = 0.30$, so a 10% variation in f_{cap} gives a 33% variation in L_{loss} — a sensitivity that underscores the importance of precisely determining f_{cap} in future simulations. Table 3 summarizes the sensitivity of L_{loss} to each input parameter.

TABLE 3 – Sensitivity of L_{loss} to input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\text{loss}} / \partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^{6} m	$\pm 3\%$	≈ 2
Ω_p	$4.99 \times 10^{-4}\ \text{s}^{-1}$	$\pm 10\%$	≈ 1
ε	$55°$	$\pm 10°$	≈ 0.5
f_{cap}	0.70	± 0.05	≈ -3

The budget is most sensitive to f_{cap} (due to the factor $1/(1 - f_{\text{cap}})$) and to R_p (due to the R_p^2 scaling law). This sensitivity motivates the priority given to Limitation L2 (numerical determination of f_{cap}).

Physical bound on L_{esc}. Independently of the exact form of the distribution, physics imposes an absolute upper bound — the same one established in Eq. 48 (Section 11.5), rederived here in non-dimensional form :

$$L_{\text{esc}} \leq M_{\text{esc}}\, r_{\text{emission}}\, \bar{v} \approx 2.25 \times 10^{22} \times 7.9 \times 10^{6} \times 9.8 \times 10^{3} \approx 1.74 \times 10^{33}\ \text{J s}. \quad (52)$$

With the nominal parameters ($f_{\text{cap}} = 0.70$, $M_{\text{ej}} = 7.5 \times 10^{22}$ kg, $I_p = 1.32 \times 10^{38}$ kg m^2, $\Omega_p = 4.99 \times 10^{-4}$ s^{-1}), the fraction of the initial angular momentum that can be carried away by escaped ejecta is at most :

$$\frac{L_{\text{esc}}}{L_{\text{init}}} \leq \frac{M_{\text{esc}}\, r_{\text{emission}}\, \bar{v}}{I_p \Omega_p} \approx \frac{1.74 \times 10^{33}}{6.59 \times 10^{34}} \approx 0.026. \quad (53)$$

With $T_{\text{init}} = 3.5$ h, the post-ejection Earth–Moon system would therefore retain at least 97.4% of the initial angular momentum, whereas the present state retains only 52.6%. The resulting deficit, $\Delta L \approx 2.95 \times 10^{34}$ J s (44.8% of L_{init}), and the minimum compatible period $T_{\text{init}} \gtrsim 6.33$ h are derived in Section 11.5 (Eqs. 49 and 50) and identified as a major limitation (Limitation L15, Section 18).

Falsifiable prediction — P21 — Numerical closure of the angular momentum budget

A Monte Carlo simulation of the ejecta spectrum (Section 11.4), with parameters sampled within their uncertainty ranges, must reproduce the accounting of the angular momentum budget (Eqs. 48 and 49, Section 11.5). The mean specific angular momentum of escaped ejecta must satisfy the bound :

$$\langle h \rangle_{\text{esc}} = \frac{L_{\text{esc}}}{M_{\text{esc}}} \leq h_{\max} = r\bar{v} \approx 7.74 \times 10^{10}\ \text{m}^2\,\text{s}^{-1}. \quad (54)$$

Given the angular spread of the distribution ($\theta_{\max} \approx 30°$, $\langle |\sin\theta| \rangle \approx 0.25$), the Monte Carlo value of L_{esc} should lie significantly *below* the bound $L_{\text{esc}}^{\max}$ — which makes the deficit ΔL (Eq. 49, Section 11.5) a conservative estimate (lower bound on the actual deficit).

Falsification : a Monte Carlo value $L_{\text{esc}} > M_{\text{esc}}\, r\, \bar{v}$ would indicate an error in the distribution $\rho(v, \theta)$ or in the accounting ; a deviation greater than $\sim 5\%$ between the Monte Carlo L_{esc} and the analytical estimate weighted by $\langle h \rangle$ would invalidate the discrete formulation of Section 11.4.

The Monte Carlo implementation of this model is simple and computationally inexpensive ; it is recommended as a first step of the 3D SPH validation program, upstream of the full hydrodynamic simulations.

12 Chronology : Moon formation in 3 to 4 weeks

12.1 The three phases of an episode

Each ejection episode comprises three phases : resonant acceleration, during which the CMT grows exponentially via elliptical instability (Section 7.6), $t_{\mathrm{acc}} = \ln(U_{\mathrm{crit}}/U_{\mathrm{turb}})/\sigma_{\mathrm{res}} \approx$ 1.4 days ; ejection, during which the flow crosses the Roche radius ($3R_{\mathrm{eq,proto}} \approx$ 22.2 Mm) at U_{crit}, $t_{\mathrm{ej}} \approx 0.6$ hour ; and recharge, during which the CMT reorganizes over 5–6 nutation periods, $t_{\mathrm{rec}} \approx$ 10–12 days. The Fe-Ni segregation rate (Stokes law) is nearly instantaneous between episodes ($\tau_{\mathrm{Stokes}} \approx 254$ s, Appendix F), which implies that each accreted layer is chemically distinct from the previous one — a direct consequence underlying seismic prediction P1.

12.2 Complete chronology

Phase	**Duration**	**Cumulative**
Episode 1 : Acceleration	1.4 days	1.4 days
Episode 1 : Ejection	0.6 h	1.4 days
Recharge (5–6 nutation periods)	10–12 days	11–13 days
Episode 2 : Acceleration	1.4 days	12–14 days
Episode 2 : Ejection	0.6 h	12–14 days
Recharge	10–12 days	22–26 days
Episode 3 : Acceleration	1.4 days	23–27 days
Episode 3 : Ejection	0.6 h	23–27 days
Total (N = 3)	**24–28 days**	

This timescale of 3 to 4 weeks is the only intermediate temporal window between the giant impact (a few hours) and a synestia ($\sim$ 100 yr) on the Hf-W chronometer (half-life : 8.9 Ma). Episode 1 (acceleration plus ejection) forms the proto-POB, $R_1 \approx$ 1,560 km. The recharge and second episode, $\approx$ 11–13 days, are asymmetric : the post-ejection transient precession modifies ε, producing the crustal dichotomy of Section 9.2. The recharge and third episode, again $\approx$ 11–13 days, reconstitute the lunar mass. The TPT window closes at $t \approx$ 30–50 Ma, and the Hadean dynamo starts at $t \approx$ 330–350 Ma.

12.3 Age of the Moon : LMO solidification versus formation

The measured age (4.425 ± 0.025 Ga) is that of lunar magma ocean (LMO) solidification, not formation. If the TPT is triggered at ≈ 4.467 Ga, the LMO cooling ($\sim$ 40 Ma) gives a predicted solidification age,

$$t_{\mathrm{LMO}} \approx 4.467 - 0.040 = 4.427\,\mathrm{Ga}, \tag{55}$$

consistent to 0.1σ with the measured value. Prediction P8 is a formation age $>$ 4.45 Ga, directly testable by Artemis III.

13 Internal stratification and seismology program

13.1 Architecture of concentric layers

Each ejected sheet originates from magma at a progressively more advanced stage of Fe-Ni segregation,

$$\left(\frac{\mathrm{Fe}}{\mathrm{Si}}\right)_{\text{layer } k} > \left(\frac{\mathrm{Fe}}{\mathrm{Si}}\right)_{\text{layer } k+1}, \tag{56}$$

because the source magma becomes progressively depleted in siderophile elements between episodes. The density contrasts between layers $\Delta\rho_{k,k+1} \approx 100\text{–}200\ \mathrm{kg\,m^{-3}}$ (Garcia et al., 2019) produce potentially detectable acoustic impedance contrasts.

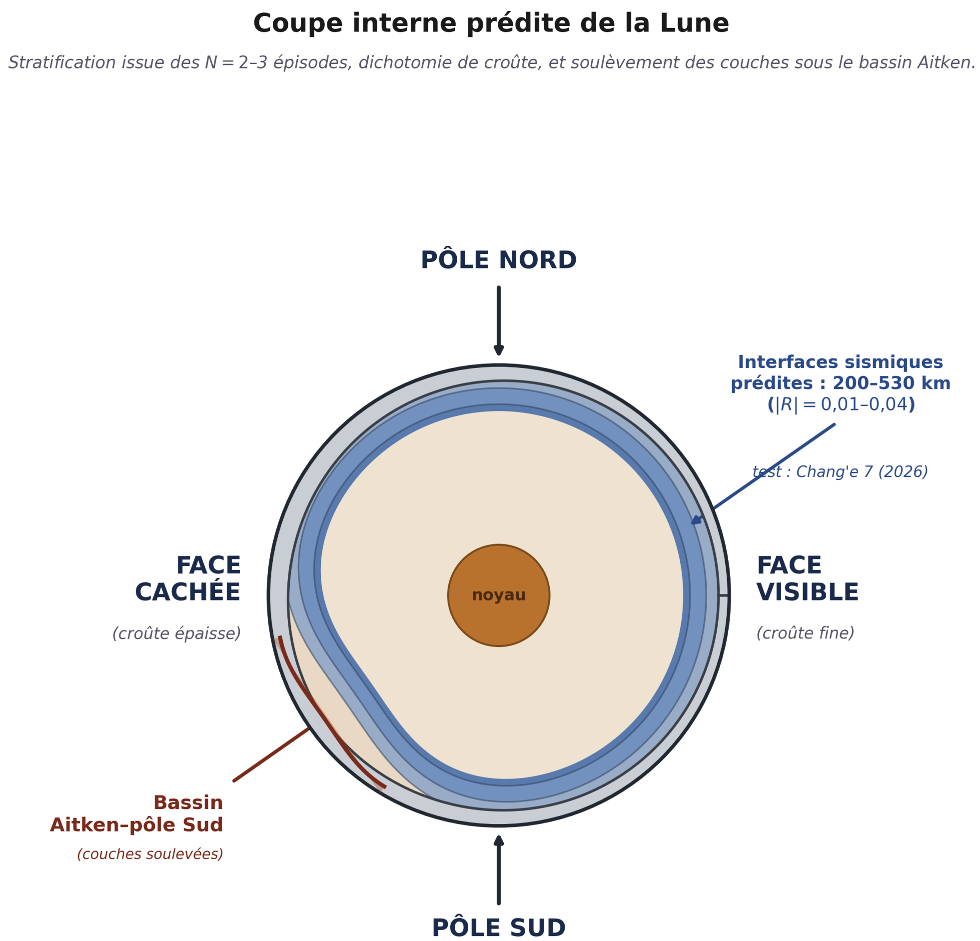

FIGURE 3 – Predicted internal cross-section of the Moon. Concentric stratification from $N = 2\text{–}3$ episodes, crustal dichotomy (thick crust on farside, thin crust on nearside), and uplift of layers under the South Pole–Aitken basin. Predicted seismic interfaces lie in the window 200–530 km ($|R| = 0.01\text{–}0.04$) ; test : Chang'e 7 (2026).

13.2 Preservation window : 200–530 km

Two destructive processes bound the preservation window. From below, the ilmenite cumulate overturn (Briaud et al., 2023) may have remelted the deepest layers (episodes 1–2, > 600 km). From above, the mare basaltic volcanism, active until ≈ 2 Ga (Li et al., 2021; Che et al., 2021), remelted the most superficial layers (< 100 km). The remaining preservation window therefore lies approximately between 200 and 530 km depth.

13.3 Interface depth : N=2 versus N=3

For $N = 2$ at equal masses, the interface between the two layers lies at :

$$d = R_{\text{Moon}} - R_1 \approx 1,737 - 1,560 \approx 177\,\text{km} \quad (N = 2). \tag{57}$$

For $N = 3$, the main interface lies in the window 200–315 km. Seismic data from Chang'e 7 will therefore discriminate between $N = 2$ and $N = 3$.

14 Observational constraints and existing evidence

14.1 Isotopic identity Δ^{17}O < 5 ppm

The Earth–Moon isotopic composition is nearly identical for O, Cr and Ti, established to better than 5 ppm (Wiechert et al., 2001; Dauphas et al., 2014; Dauphas, 2017; Sossi et al., 2025). The TPT satisfies this constraint *mechanically* : at each ejection episode, the matter is drawn from the contemporary Hadean Earth's mantle, with no exogenous isotopic source. No *ad hoc* isotopic mixing is required.

14.2 Small lunar core

The Moon has a metallic core of radius ≈ 330 km, significantly smaller than Earth's core relative to the host planet's mass (O'Neill, 1991; Jones & Drake, 1986). In the TPT, this reduction is a direct consequence of the unique engine : at each successive episode, the ejected mantle is more depleted in siderophiles (metallic Fe-Ni) than the previous one. The remaining iron in the lunar mantle is mainly in the form of FeO, whose concentration is indeed higher than in the terrestrial mantle (Section 14.4).

14.3 Angular momentum

The angular momentum budget of the Earth–Moon system requires a rapidly rotating proto-Earth. The value $T_{\text{rot}} = 3.5$ h is consistent with the median value of N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010) : rapid rotation is the norm at the end of accretion, not an exception.

14.4 Volatiles and FeO : two additional constraints

Volatile depletion. The Moon is strongly depleted in volatile elements (K, Na, Zn) compared to Earth and chondrites, with isotopic fractionation indicating high-temperature devolatilization (Humayun & Clayton, 1995; Paniello et al., 2012; Wang & Jacobsen, 2016; Dai et al., 2025). A cohesive ejection with a protective glassy crust might seem to minimize this loss. However, the ejected magma is at $T \approx 3,000$–$4,000$ K before crust formation, and devolatilization occurs during ballistic expansion, on timescales ($\sim 10^2$–10^3 s) comparable to the superficial cooling time. The TPT therefore predicts a selective volatile depletion, although the exact quantification depends on the surface-to-volume ratio of the ejected tongue and the vapor pressure of the species considered.

Mantle FeO. The Moon is depleted in *metallic iron* (small core), but the FeO in its mantle is higher ($\approx$ 13% by weight) than in the Earth's mantle ($\approx$ 8%) (Taylor et al., 2006; Khan et al., 2006; Garcia et al., 2019). Fe-Ni segregation does not affect FeO ; it concerns metallic iron. The FeO of the lunar mantle reflects the composition of the Hadean Earth's mantle before segregation, or an enrichment by fractional crystallization of the lunar magma ocean. The TPT has no quantitative prediction for FeO. To be complete, a future version of the model will need to quantify the evolution of FeO through the ejection episodes and the LMO cooling.

14.5 Chang'e-6 results : preliminary observational support

The Chang'e-6 mission returned 1,935 g of samples from the Apollo basin, within the South Pole–Aitken (SPA) region, in June 2024. Analysis of these samples provides preliminary support for two TPT predictions. Yue et al. (2026) dated Chang'e-6 norite samples to 4247 ± 5 Ma by the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt sheet, representing deep lunar mantle material excavated by the SPA impact. This age is consistent with the TPT prediction that the innermost lunar layer (Episode 1) was accreted at $\approx$ 4.5 Ga and partially reset by the SPA impact at $\approx$ 4.25 Ga. Xu et al. (2025) further report olivine fragments in the Chang'e-6 samples containing Fe and Mn concentrations significantly higher than those found in the standard lunar mantle, with zinc concentrations $\approx 100\times$ higher than expected. Although part of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed by mixing of lunar rocks and melted asteroidal material. The high iron content (total Fe, including FeO) of the material excavated from SPA depths is *consistent* with the Fe/Si gradient increasing with depth (P3) and provides preliminary support for the South Pole Fe/Si anomaly (P6), detailed below.

15 Falsifiable predictions

This section gathers, in catalog form, each quantitative and falsifiable prediction made by the TPT. Each entry states the prediction, its physical derivation when not already fully given elsewhere in the manuscript, the relevant test, and an explicit falsification threshold set in advance of observational data.

Falsifiable prediction — P1 — Seismic interface at $d \approx 200$–315 km (Priority 1)

The concentric stratification produces acoustic impedance contrasts $|R| \in [0.01, 0.04]$ between layers. For $N = 2$, the interface lies at $d \approx 177$ km. For $N = 3$, the main interface lies at $d \approx 200$–315 km.

Detectability with Chang'e 7 depends on the seismometer sensitivity (target 10^{-8} m s^{-2} Hz$^{-1/2}$ at 0.1–1 Hz). This sensitivity should resolve $|R| > 0.01$, provided at least 10 $M > 2.5$ events are recorded during the first 6 months of operation.

Falsification : no phase conversion detected within the window 200–530 km after deployment of at least 3 seismic stations and detection of at least 10 lunar events of $M > 2.5$.

Missions : Chang'e 7 (August 2026, South Pole seismometer), FSS (Farside Seismic Suite), LEMS (Lunar Environment Monitoring Station), Artemis III (2028–2029).

Falsifiable prediction — P2 — Seismic asymmetry, nearside versus farside

The second, asymmetric episode produces a sharper and deeper interface beneath the farside.

Falsification : spherical symmetry confirmed by a complete tomographic inversion.

Falsifiable prediction — P3 — Fe/Si gradient increasing with depth

The Fe/Si ratio increases with depth in the lunar mantle, reflecting the progressive Fe-Ni depletion of the source magma between episodes. Quantitatively, with the total cumulative fraction of Fe removed from the mantle $\xi_{\mathrm{Fe}}^{\mathrm{sat}} \approx 0.15$ (Rubie et al., 2015) and $N = 3$ segregation episodes, the fraction of Fe removed per episode, assuming geometric progression over the N episodes, is :

$$\xi_{\mathrm{Fe,episode}} = 1 - (1 - \xi_{\mathrm{Fe}}^{\mathrm{sat}})^{1/N} \approx 1 - (0.85)^{1/3} \approx 0.053, \tag{58}$$

i.e., about 5.3% of the remaining Fe removed at each episode. The resulting Fe/Si ratio between adjacent layers is multiplicative :

$$\frac{(\mathrm{Fe/Si})_{\mathrm{Ep}k}}{(\mathrm{Fe/Si})_{\mathrm{Ep}(k+1)}} = \frac{1}{1 - \xi_{\mathrm{Fe,episode}}} \approx 1.056, \tag{59}$$

giving an enrichment of about 5.6% between Episode 2 and Episode 3, and a cumulative enrichment $(\mathrm{Fe/Si})_{\mathrm{Ep1}}/(\mathrm{Fe/Si})_{\mathrm{Ep3}} = (1 - \xi_{\mathrm{Fe,episode}})^{-2} \approx 1.115$, i.e., a total enrichment of about 11.5% between Episode 1 (the deepest, most Fe-rich layer) and

Episode 3 (the most superficial). The SPA basin preferentially exposes Episode 1 material.

Preliminary support : Chang'e-6 olivines with high Fe/Mn ratios (Xu et al., 2025). **Missions :** Chang'e-6 (additional analyses), Artemis III.

Falsifiable prediction — P4 — Dynamo delay : 290–360 Ma

The progressive emergence of the dynamo predicts a continuous growth of paleointensity from 4.55 to 4.20 Ga, without abrupt onset (Tarduno et al., 2025).

Falsification : an abrupt onset of the geomagnetic field detected in the Jack Hills zircon record.

Falsifiable prediction — P5 — Instability window : $\varepsilon \in [57°, 70°]$

The parametric elliptical instability requires ε within this interval.

Test : N-body simulations of terrestrial protoplanet obliquity.

Falsifiable prediction — P6 — Fe/Si anomaly at the South Pole (derived from P3 and the South Pole–Aitken impact simulations of Wakita et al. 2026)

Observational context. The hydrodynamic simulations of Wakita et al. (2026) establish three facts independent of the TPT : the SPA impact excavated lunar mantle material from depths greater than 90 km ; the oblique, north-to-south trajectory of the impactor produced an asymmetric ejecta plume that preferentially concentrated deep material toward the South Pole ; and about 350 m of mantle material is predicted at the surface near Shackleton crater on average.

TPT interpretation. Within the TPT framework, the lunar mantle is stratified into concentric layers (P1), with Fe/Si decreasing from the deepest layer (Episode 1) to the most superficial (Episode 3), as quantitatively derived in P3 above. The combined thickness of Episode 2 and Episode 3 layers is about 315 km (Section 13). The material excavated by the SPA impact from depths greater than 90 km, and concentrated toward the South Pole, therefore comes preferentially from Episode 2 and, at greater excavation depths, from Episode 1.

Quantitative scenario. Wakita et al. (2026) do not provide a quantified fraction of Episode 1 versus Episode 2 material in the SPA ejecta plume, since their simulation does not incorporate the TPT stratigraphy ; this fraction must therefore be bounded from first physical principles rather than read directly from their results. We consider the most unfavorable scenario compatible with the TPT : the SPA excavation reaches only $\approx$ 100 km (barely beyond the 90 km minimum established by Wakita et al. 2026), so that only Episode 2 material — and not the deeper, more strongly enriched Episode 1 — contributes to the ejecta, and we conservatively take the minimum plausible fraction of Episode 2 material in the plume as 30%. With the Fe/Si enrichment of about 5.6% of Episode 2 relative to Episode 3 derived in Eq. 59, this minimal scenario predicts an enrichment of :

$$\Delta(\mathrm{Fe/Si})_{\mathrm{min}} \approx 0.30 \times 0.056 \approx 0.017 \approx 1.7\%. \tag{60}$$

If the excavation instead reaches more than about 315 km, Episode 1 material contributes directly ; as an illustration, a mixture of 30% Episode 1, 30% Episode 2, and 40% Episode 3 material would give $\Delta(\mathrm{Fe/Si}) \approx 0.30 \times 0.115 + 0.30 \times 0.056 \approx 5.1\%$, a value that requires a non-negligible Episode 1 contribution to be reached.

Quantitative prediction. Samples from the South Pole (Chang'e-7, Artemis III) should show a higher Fe/Si ratio than the average of equatorial (Apollo) and nearside (Chang'e-5) samples, with an enrichment of at least approximately 1.7% in the minimal scenario above and potentially up to about 11.5% if Episode 1 material dominates the sampled ejecta.

Falsification threshold. We set, in advance of any observational data, the falsification threshold at :

$$(\mathrm{Fe/Si})_{\mathrm{SPA}} \leq 1.015 \times (\mathrm{Fe/Si})_{\mathrm{average\,lunar}} \implies \text{P6 falsified,} \tag{61}$$

i.e., an enrichment below 1.5%, set slightly below the minimal scenario value of 1.7% from Eq. 60 to account for experimental uncertainty without weakening the essence of the test : even the most conservative scenario compatible with the TPT predicts an enrichment above this threshold, so a measured enrichment below it would be incompatible with the theory as constructed here. An enrichment between 1.5% and 5.6% (the maximum achievable by Episode 2 material alone, from Eq. 59 applied to a pure Episode 2 sample) is read as partial support, consistent with ejecta dominated by Episode 2 without requiring Episode 1 contribution. An enrichment above 5.6% cannot be produced by Episode 2 and 3 material alone, and therefore constitutes direct evidence for Episode 1 contribution, providing strong support for the deep stratification predicted by P1 and P3.

Epistemic note on the gradation bounds. The lower bound (1.5%) rests on an estimate of the minimum fraction of Episode 2 material in the SPA ejecta (30%), which is a modeling assumption. The upper bound (5.6%) is a physical constraint : it is the maximum enrichment achievable by a pure Episode 2–Episode 3 mixture, without Episode 1 contribution. Consequently, any measured enrichment above 5.6% constitutes definitive evidence for the presence of Episode 1 material, a qualitatively different regime from the lower-bound scenario. The two bounds are therefore not of the same epistemic nature ; this asymmetry is explicitly recognized here.

Test : Chang'e-7 surface analyzers and samples returned by Artemis III from the South Pole, compared to existing Fe/Si measurements from Apollo (equatorial) and Chang'e-5 (nearside).

Recognized limitation : this prediction shares Limitation L9 (impact mixing in SPA ejecta) with prediction P3, and Limitation L11 (the 30% Episode 2 fraction itself) below.

Falsifiable prediction — P7 — Signature in the Hf-W chronometer

A formation duration of 3 to 4 weeks is the only intermediate temporal window between the giant impact (a few hours) and a synestia ($\sim$ 100 yr) on the Hf-W chronometer (half-life : 8.9 Ma). The ^{182}Hf/^{180}Hf ratio in Episode 1 samples should correspond to a differentiation age of 4.467 Ga (100 $\pm$ 10 Ma after CAIs), compared to 60 $\pm$ 10 Ma for the giant impact (Kleine et al., 2002).

Mission : Artemis III (2028–2029).

Falsifiable prediction — P8 — Formation age > 4.45 Ga

The measured age of 4.425 Ga is that of LMO solidification, not formation (Section 12). The formation age is about 4.467 Ga.

Test : direct dating of Episode 1 rocks exposed on the central peaks of the SPA basin (Artemis III).

Falsifiable prediction — P23 — Magma rheology constrained by South Pole–Aitken basin samples

The South Pole–Aitken basin plays the role of a window for the TPT : the impact — posterior to formation — thinned the crust and uplifted the deep Episode 1 layers (Figure 3), exposing mafic materials (olivine, magnesian pyroxenes) of deep origin at the basin center, whereas the periphery and rings show only reworked crust. It is therefore the basin center that constitutes the sampling target for this prediction.

The target measurement is not textural but compositional : the textures of the samples record the SPA impact and its cooling, not the Hadean emplacement. The bridge to the primordial rheology passes through the laboratory : re-melting of representative compositions and measurement of the laws $\eta(\phi, T)$ and $\tau_y(\phi, T)$ — viscosity and yield stress as a function of crystallinity and temperature (Caricchi et al., 2007; Giordano et al., 2008) — subsequently applied to the thermal state predicted by the TPT. The samples already exist in part (Chang'e-6, basin rim) and will be supplemented by Chang'e-7 and Artemis III.

The stakes of this single measurement are twofold, for the same parameter τ_y governs two independent clocks of the model : the cohesion of the ejected tongue ($\mathrm{Bi}_{\mathrm{KH}}$, Appendix D, Limitation L14) and the speed of lunar tidal recession toward the evection resonance ($Q \propto 1/\tau_y$, Section 11.5, Limitation L15). A single measurable quantity thus simultaneously constrains the two highest-priority limitations of the theory. We also note, with caution, that the deep mass anomaly detected beneath the basin (James et al., 2019) is compatible with the uplift of dense layers predicted by Figure 3, without demonstrating it — the standard interpretation (buried impactor core) remaining admissible.

Falsification : if the rheology measured at the TPT thermal state gives τ_y outside the range 10^2–10^4 Pa, the cohesion ($\mathrm{Bi_{KH}}$, Appendix D) and recession (Section 11.5) windows must be recalculated ; a τ_y incompatible with $\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} \geq 1$ after recalculation would refute the cohesive ejection mechanism.

Missions : Chang'e-6 (samples in hand), Chang'e-7, Artemis III.

Falsifiable prediction — P9 — Compatibility of mare basalts with Layer 1

The melting depth of mare basalts (200–500 km, ± 100 km) is compatible with a source within Layer 1 ($> d_{\mathrm{P1}}$), although this cannot be asserted in the absence of more precise geochemical data.

Test : high-resolution isotopic geochemistry of Apollo and Chang'e-5 mare basalts (Li et al., 2021).

Falsifiable prediction — P10 — Seismic interface at $d \approx 177$ km for $N = 2$

For $N = 2$ episodes of equal mass, the single interface between the two concentric layers lies at $d \approx 177$ km. This depth lies outside the main preservation window (200–530 km) because it was partially remelted by mare volcanism and/or ilmenite cumulate overturn. However, partial remnants could be detected as weak reflectors or seismic velocity variations in the 150–200 km zone.

Predictions P17 (capture efficiency, Section 10.3), P20 (asymptotic regime indicator, Section 11.6), P21 (numerical closure of the angular momentum budget, Section 11.5), and P22 (the unified ejection threshold, Section 9.2) are derived in full, with their own falsification conditions, at the manuscript points indicated above, rather than repeated here.

16 Validation windows : three imminent tests

16.1 Chang'e 7 : imminent seismic validation (August 2026)

The Chang'e 7 mission (CNSA, launch planned for August 2026) will deploy the first seismometer at the lunar South Pole since Apollo 17 (1972). Its data will provide a direct test of P1, P2 and P6. The prediction is pre-registered and quantified : $d \approx 200$–315 km, $|R| \in [0.01, 0.04]$ for the seismic interface, and an Fe/Si enrichment above 1.5% for South Pole ejecta. The falsification criteria are explicit in each case.

16.2 The South Pole–Aitken basin : stratigraphic revealer

The SPA basin did not create the concentric lunar stratigraphy : it *excavated* it. The layer structure pre-existed the SPA impact at ≈ 4.25 Ga (Yue et al., 2026), encoded during TPT differentiation at ≈ 4.5 Ga. The impact is the accidental revealer of a much older internal architecture.

High-resolution 3D simulations (Wakita et al., 2026) show that the north-to-south oblique trajectory of the SPA impactor preferentially launched deep mantle ejecta toward the South Pole in a butterfly-shaped plume — precisely the landing zone of Artemis III. These same simulations predict about 350 m of average mantle material at this site, excavated from depths greater than 90 km, consistent with both the TPT preservation window and the quantitative scenario underlying prediction P6.

16.3 Artemis III : direct sampling of Episode 1 material

The Artemis III mission (NASA, planned for 2028–2029) will land astronauts near Shackleton crater at the South Pole. According to Wakita et al. (2026), surface samples at this location could include deep mantle ejecta from the SPA impact. If so, these samples will enable a direct test of P3, P6, P7 and P8 : their Fe/Si ratio, crystallization age ($\approx$ 4.467 Ga for primary Episode 1 material), and isotopic signature ($\Delta^{17}O$ $<$ 5 ppm relative to the contemporary Earth's mantle) are all quantitative predictions of the TPT.

Chang'e 7 (2026) targets seismology and the Fe/Si anomaly (P1, P2, P6), while Artemis III (2028) targets geochemistry and chronology (P3, P6, P7, P8). Two independent space agencies converge on a unique geographic site — the lunar South Pole — to test a single internal architecture.

17 Discussion : main objections and responses

17.1 Comparison with the giant-impact model

Constraint	Giant impact	TPT (this work)
Isotopic identity	Requires very specific impactor composition and mixing conditions	Mechanically satisfied by construction
Small lunar core	Requires a pre-differentiated impactor (Theia)	Direct consequence of the unique engine
Crustal dichotomy	Not naturally predicted	Qualitative prediction : post-Episode 2 precession (L7)
Dynamo delay	Not predicted	Proposed : $\approx$ 350 Ma, no free parameter (L4)
Seismic predictions	None	P1 testable by Chang'e 7 (August 2026)
Chang'e-6 Fe anomaly	Not predicted	Consistent with P3 and P6 (Xu et al., 2025; Yue et al., 2026)
Formation timescale	A few hours	3–4 weeks
Hf-W signature	60 ± 10 Ma after CAIs	100 ± 10 Ma after CAIs (P7)

A recent review concludes that there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in the formation of the Moon (Sossi et al., 2025). This conclusion motivates the exploration of alternatives based on the internal dynamics of the proto-Earth.

17.2 Responses to main objections

Objection 1 : "A large number of unjustified hypotheses." The hypotheses are explicitly classified into three levels (Section 8, Table 2) : Level 1 (established) designates results from the literature or closed-form derivations (e.g., $E_{\text{grav}}/E_{\text{fus}} \approx 121$, the Bingham-Herschel law) ; Level 2 (constrained) designates physically motivated hypotheses not yet numerically validated (e.g., $\varepsilon \in [40°, 70°]$) ; Level 3 (speculative) designates hypotheses awaiting validation by 3D SPH (e.g., the emergence of a single $\ell = 1, m = 1$ torus, the exact value of f_{cap}, the quantitative validity of the Gaussian approximation discussed below). The TPT does not rest on all these hypotheses with equal weight. The observational predictions (Section 15) are independent of the Level 3 hypotheses.

Objection 2 : "The CMT has not been demonstrated." In the rapid rotation limit (Ro $\ll$ 1), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential vorticity conservation equation. Linearizing about the mean flow near the instantaneous dynamical equator ($\phi = 0$, perpendicular to $\mathbf{\Omega}(t)$, as defined in Section 3), and applying the standard multi-scale expansion for Rossby wave envelopes (Redekopp, 1977; Luo, 1991; Yin et al., 2019), one obtains a nonlinear Schrödinger equation for the envelope $A(y,t)$, where $V_{\text{eff}}(y)$ is the sum of the potentials listed in Table 1. A stationary ansatz $A(y,t) = \psi(y)e^{-iEt/\hbar_{\text{eff}}}$ then gives a time-independent nonlinear Schrödinger equation for $\psi(y)$, the nonlinearity arising from the advective term of the potential vorticity equation. The fundamental mode of the *linearized* version of this equation, with $V_{\text{eff}}(y)$ expanded to quadratic order about $y = 0$, is a Gaussian wavefunction centered on the instantaneous equator, confined to the belt $|\phi| < 30^\circ$ — a direct consequence of the fact that all seven potentials of Table 1 have their extrema at $y = 0$. Whether this Gaussian remains a quantitatively adequate approximation to the ground state of the full nonlinear equation, rather than merely its first-order qualitative limit, has not been established for this system and is recognized as Limitation L12 (Section 18). The complete derivation is given in Appendix C. Quantitative validation of the single-torus structure awaits 3D SPH simulation (Limitation L1).

Objection 3 : "The ejection mechanism relies on an adjusted threshold." The ejection threshold is defined by the condition $\lambda(U) = 0$ (Section 9.2), a deterministic surface in parameter space. It is a function of the density gradient β, the obliquity ε, and the angular speed Ω : ejection occurs when $\Omega > \Omega_{\text{crit}}(\varepsilon, \beta)$. The number of episodes $N = 2$–3 results from the equilibrium between the recharge rate (fueled by angular momentum input) and the loss by ejection. It is a consequence of the Hadean conditions (β, momentum input), themselves constrained by the existence of the Moon — a retrodictive result, not a free parameter.

Objection 4 : "$f_{\text{cap}} = 0.70$ is high compared to accretion simulations (0.3–0.5)." The classical range (0.3–0.5) concerns impacts between solid bodies. Here, the ejected material is a magma at $3,000$–$4,000$ K, with viscous coalescence and toroidal orbital confinement. Appendix G establishes a physical lower bound $f_{\text{cap}}^{\min} \gtrsim 0.45$–$0.55$, independent of the BH cohesion assumption. The value 0.70 lies within a mechanically justifiable range.

Objection 5 : "Are the required thresholds for ejection physically reachable ?" The ejection threshold is a condition on Ω : $\Omega > \Omega_{\text{crit}}$, with $\Omega_{\text{crit}} \approx \sqrt{2}\,\Omega_0$. This factor is reachable by the angular momentum input from continuous planetesimal bombardment (Appendix H). The Euler oscillation term λ_4 modulates the effective obliquity ε, but does not trigger ejection directly.

Objection 6 : "The manuscript is too speculative for a review." The TPT is submitted as a *theory with falsifiable predictions*, and not as an observational result. Validation will proceed in two phases : Phase 1 (2026–2029), observational tests by Chang'e 7 (August 2026) and Artemis III (2028–2029) ; and Phase 2 (2027–2030), 3D SPH numerical validation once institutional resources are committed. The TPT will be judged on its predictions, and not on a prior simulation.

17.3 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile : differential equatorial-pole cooling generates a density gradient and thermal acceleration of rotation ; Fe-Ni segregation lightens the CMT, and by angular momentum conservation, its inner part accelerates. The Bingham-Herschel threshold also blocks small scales, favoring a block-like profile. Numerical validation of this profile is a priority for 3D SPH simulation.

17.4 Partial fragmentation and coalescence

Even if the ejected sheet partially fragments, the fragments remain within the same orbital zone and coalesce rapidly under inelastic collisions and gravitational attraction of the proto-POB (Canup & Asphaug, 2001). The effective capture yield therefore remains high ($f_{\text{cap}} \geq 0.7$). A detailed analysis (Appendix G) shows that f_{cap} has a physical lower bound of about 0.45–0.55, independent of the BH cohesion assumption.

18 Recognized limitations

A 3D SPH simulation is necessary, but premature at the time of writing (July 2026). Such a simulation — coupling the Bingham-Herschel rheology, radiative cooling, Fe-Ni segregation, and the geometry $\varepsilon \neq 0$ — requires institutional HPC resources that are not committed at this stage, for a simple reason : the first decisive observational test arrives before such a simulation could reasonably be completed. Chang'e 7 is scheduled for August 2026, only a few weeks away. Its seismic data will detect, or not detect, the predicted interface within the window 200–530 km (Prediction P1). In the former case, the concentric stratification predicted by the TPT receives independent observational support, and a dedicated 3D SPH validation becomes a justified investment of institutional resources. In the latter case, under the falsification conditions stated in Section 15, the central prediction of the TPT is seriously undermined, and an expensive numerical campaign built on a falsified premise would be of little value. This is why the present work is submitted, and should be evaluated, on the basis of its falsifiable observational predictions, upstream and independently of any numerical validation campaign.

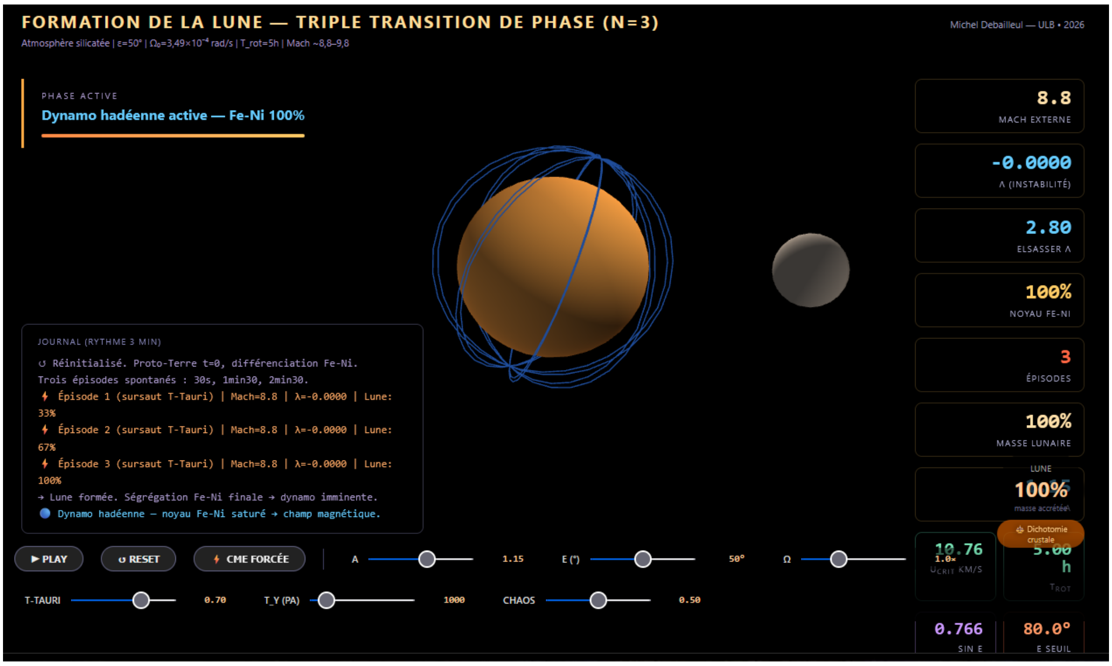

FIGURE 4 – Screenshot of the interactive simulation accompanying this work (link above), illustrating the Coherent Magmatic Torus (CMT) during ejection around the proto-Earth, with the real-time dynamic dashboard of parameters (external Mach number, instability λ, Elsasser number, Fe-Ni core fraction, accreted lunar mass).

Recognized limitation — L1 — Spontaneous organization of the CMT

The question of whether a single torus or multiple coexisting structures emerge in the exact Hadean regime can only be resolved by a high-resolution 3D SPH simulation with full BH rheology. **Priority 2.**

Recognized limitation — L2 — Numerical value of f_{cap}

The capture efficiency $f_{cap} = 0.70$ is physically motivated but not derived from first principles. Its increasing dependence on M_{Moon} and its deviation from the classical disk accretion range (10–55%, Kokubo et al. 2000) are formalized under the testable prediction P17. 3D SPH simulation remains the ultimate arbiter of the exact value of f_{cap}.

Recognized limitation — L3 — Small-scale cohesion

The $\mathrm{Bi}_{KH} \gg 1$ argument is valid at large and intermediate scales. At very small scales ($\lambda_{KH} \ll R_{torus}$), partial fragmentation cannot be excluded analytically. However, fragments from the same toroidal flow, ejected with the same speed and angle, end up on nearly identical orbits and accrete onto the POB : the mass budget is not affected by fragmentation, only the fragment size is.

Recognized limitation — L4 — Dynamo delay : ± 50 Ma

The exponential core growth model is simplified ; the associated uncertainty is

± 50 Ma.

Recognized limitation — L5 — Super-rigid profile $\alpha > 1$

The super-rigid rotation profile ($U \propto r^{\alpha}$, $\alpha > 1$) is physically motivated by differential rotation in a partially crystallized magma undergoing rapid cooling, but has not been demonstrated under these exact Hadean conditions. It is a Level 3 hypothesis, to be validated by simulation.

Recognized limitation — L6 — Flow $\rightarrow$ POB transition : qualitative status

The three-phase description of the ejected flow (ballistic expansion, orbital insertion, cohesive accretion beyond R_{Roche}) is physically motivated and consistent with the calculated order of magnitudes, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required. **Priority 2b.**

Recognized limitation — L7 — Crustal dichotomy : qualitative prediction

The thickness ratio $\approx 2{:}1$ is a qualitative prediction of the TPT. The coupled differential thermal calculation has not yet been derived analytically. The lunar farside underwent at least two independent remelting episodes that must be accounted for in any quantitative model : the SPA basin-forming impact at ≈ 4.25 Ga, which excavated and partially remelted the deepest Episode 1 layer on the farside (Wakita et al., 2026) ; and prolonged mare volcanism extending until ≈ 3.2 Ga (Li et al., 2021), which remelted the most superficial layers (< 100 km) over much of the farside. The net effect is to deepen and strengthen the existing asymmetry rather than erase it.

Recognized limitation — L8 — Rossby wave speed in a BH fluid

The rigorous derivation of $c^{(\mathrm{BH})}_{\mathrm{Rossby}}$ for a BH fluid on an oblate spheroid ($\varepsilon \neq 0$) remains an open problem. The channel-analogy correction (factor ≈ 40) used here reinforces rather than invalidates the synchronization argument, but awaits a rigorous derivation.

Recognized limitation — L9 — Impact mixing in SPA ejecta

The geochemical signature of the deep mantle material excavated by the SPA impact is potentially diluted or altered by mixing with impactor material and local crust. Quantitative deconvolution of the primary Episode 1 signature from the impact melt breccia requires detailed petrological and isotopic studies. This limitation affects predictions P3 and P6.

Recognized limitation — L10 — Deep mantle samples predicted but not yet collected

The TPT predicts that deep Episode 1 mantle material is exposed at the central peaks of craters within the South Pole–Aitken basin, excavated and extruded during the SPA impact at ≈ 4.25 Ga (Wakita et al., 2026). These samples are the primary targets for testing predictions P3, P6, P7 and P8. No samples from these central peaks have yet been collected or analyzed. The absence of deep mantle material at the predicted locations, or an Fe/Si ratio incompatible with P3 or P6, would seriously question the TPT.

Recognized limitation — L11 — Fraction of Episode 2 material in SPA ejecta

Prediction P6 requires an estimate of the minimum fraction of Episode 2 material present in the SPA ejecta plume reaching the South Pole, which we have conservatively set at 30% based on the excavation-depth argument of Section 15 rather than on a quantitative direct result of Wakita et al. (2026), whose simulations were not designed to track the specific TPT stratigraphic layers. A future dedicated hydrodynamic simulation, tracking the depth of provenance of SPA ejecta material, would allow refinement of this fraction — and hence of the falsification threshold of Eq. 61.

Recognized limitation — L12 — Quantitative validity of the Gaussian approximation for the CMT confinement mode

The nonlinear Schrödinger equation derived in Section 17 (Objection 2) and Appendix C is not solved exactly in this work. The Gaussian wavefunction is a reasonable approximation for the fundamental mode, valid in the limit where the wave amplitude is small, i.e., when the perturbation velocity δU associated with the $\ell = 1, m = 1$ mode satisfies $\delta U \ll U_{\text{turb}}$. This condition is qualitatively consistent with the Rhines inverse cascade (Section 5), but its exact quantitative verification is not possible with the parameters available at this stage : no independent estimate of $\delta U / U_{\text{turb}}$, nor of the nonlinear coefficient β in the governing equation, has been derived for this specific system, unlike the classical atmospheric and oceanic Rossby wave systems for which the reduction was originally established (Redekopp, 1977; Luo, 1991; Yin et al., 2019). The absence of this verification is recognized here as a limitation. The Gaussian solution should therefore be understood as the first-order approximation of the linearized problem, and not as an exact or quantitatively bounded solution of the full nonlinear equation.

Recognized limitation — L13 — Persistence of post-ejection oscillation

The mechanism of post-shock nutation persistence by reflection at the boundaries of the ejection cavity (Appendix I) relies on an analogy with established results for inertial waves in a confined homogeneous fluid sphere (Zhang et al., 2001; Aldridge & Lumb, 1987), and not on a derivation specific to the oblique geometry and Bingham-Herschel rheology of the Hadean proto-Earth. The plateau duration $T_{\text{nut}}^{(\text{post})}$ and the absence of significant decay during this window are therefore qualitative estimates, and not derived results. A full quantitative formulation — treating the coupled system $(\Omega, \varepsilon, |g_{\text{eff}}|)$ as an underdamped Langevin dynamics, integrable numerically by a BAOAB scheme (Leimkuhler & Matthews, 2013) once the physical noise (CMEs, impacts) is properly characterized as a jump process rather than Gaussian white noise — is identified as a future computational program, in complement to the 3D SPH simulation (Limitations L1, L2).

Recognized limitation — L14 — Quantification of $\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}}$ and numerical validation

The multiplicative product of the three cohesion reinforcements ($f_{\mathrm{atm}} \times f_{\mathrm{cryst}} \times f_{\mathrm{RT}}$) is established qualitatively. A 3D SPH simulation is necessary to validate the exact numerical values, in particular the nonlinear interaction between atmospheric confinement, crystal network, and Rayleigh-Taylor suppression effects. **Priority 2c.**

Recognized limitation — L15 — Angular momentum budget and required external sink (Priority 1)

The angular momentum budget reads $L_{\mathrm{init}} = L_{\mathrm{TL}} + L_{\mathrm{esc}} + L_{\mathrm{tidal}} + L_{\mathrm{autres}}$, where $L_{\mathrm{esc}} \leq 1.74 \times 10^{33}$ J s is the conservative bound (pre-shock speed, Section 11.5) carried by escaped ejecta — an upper bound, not a central estimate ; the effective post-shock speed is likely lower, which only reinforces the following conclusion rather than weakening it. This bound covers only a minor fraction of the deficit $\Delta L \approx 2.95 \times 10^{34}$ J s. Most of the deficit must therefore be evacuated by an external mechanism.

The evection resonance (Ćuk & Stewart, 2012; Ćuk et al., 2016) is the most studied quantified candidate, but its efficiency for $Q \sim 10^4$ (the "near-totally molten" state motivated in Section 4.1) is insufficient in the current state of the calculation (crossing time $\sim 3 \times 10^3$ yr vs required $\sim 5 \times 10^4$ yr) ; only $Q \gtrsim 10^5$–10^6 offers sufficient margin, which constrains the required thermal state toward a regime closer to pure fluid than to "near-totally molten" in the sense of Henning et al. (2009). Quantitative confirmation of this mechanism, and identification of any additional channels, requires a dedicated simulation following Rufu & Canup (2020).

In the current state of the manuscript, the angular momentum budget is not closed for the nominal value of Q associated with $\phi = 0.1$–0.3. This tension is identified as the major limitation of the TPT in its present version.

19 Conclusion

This work proposes that the formation of the Moon could be the product of a Triple Phase Transition within a fully molten proto-Earth, rotating at ≈ 3.5 h, whose axis oscillates within $[40°, 70°]$ in its own co-rotating reference frame — an interval supported by five convergent arguments, the first and most fundamental being the logically necessary absence of any tidal stabilizer. A unique engine — the progressive segregation of Fe-Ni — simultaneously governs three transitions : rheological, mechanical and magnetic. The result is a Moon built layer by layer in two to three massive episodes, each layer archiving in its internal structure the state of Hadean differentiation at the time of its accretion.

The parametric elliptical instability — established in fluid mechanics (Malkus, 1968; Kerswell, 2002; Lacaze et al., 2004) but never applied to lunar formation — amplifies the CMT to the bifurcation speed in about 37 hours. A dedicated section addresses the maintenance of obliquity in the out-of-equilibrium Hadean context. A chronology of three to four weeks gives rise to several new testable predictions.

The mass budget is consistent without forcing : the target mass for the TPT mechanism is less than the presently observed lunar mass, because the SPA impactor and late accretion contributed subsequently. At $T_{\rm rot} = 3.5$ h, the ejectable mass per episode can reach about 70% of the present lunar mass, which explains why $N = 2$–3 episodes suffice.

The question of the angular momentum budget, however, remains open : a post-formation external sink of about 2.95×10^{34} J s is required. The evection resonance and the Laplace plane transition (Ćuk & Stewart, 2012; Ćuk et al., 2016) provide a quantified candidate — precisely requiring the high obliquity predicted by the TPT and accounting for the present Earth's obliquity and lunar inclination — whose efficiency for the TPT parameters must be validated by a dedicated tidal evolution calculation (Limitation L15).

Preliminary observational support comes from Chang'e-6 : norites at 4247 ± 5 Ma (Yue et al., 2026) and high-Fe/Mn olivines (Xu et al., 2025) are consistent with predictions P3 and P6, although not yet confirmatory. This edition further provides, for the first time, a fully quantified version of P6 : a pre-registered threshold of 1.5% Fe/Si enrichment for South Pole ejecta, derived from first principles and not adjusted to match existing data, with explicit gradation distinguishing partial support (Episode 2 material alone) from strong support (an unambiguous Episode 1 signature).

The central prediction remains : one or more seismic interfaces within the window 200–530 km (or at ≈ 177 km for $N = 2$), testable by Chang'e 7 in August 2026. If an interface with $|R| > 0.02$ is detected within this window, the concentric stratification receives its first strong observational support, upstream of any SPH simulation. If no interface is detected under the falsification conditions stated, prediction P1 is seriously questioned and the theory will require revision — which the author explicitly accepts.

This work invites the planetary science community to evaluate the TPT on its own terms : the equations are numbered, the derivations are explicit, the hypotheses are hierarchized, the limitations are recognized, and the predictions are falsifiable. The TPT will be judged primarily on its observational predictions — the seismic data from Chang'e 7 (August 2026) and the samples from Artemis III (2028–2029) — before and independently of any full numerical validation. The theory stands or falls on these results, and not on the elaboration presented in this manuscript.

A Adiabatic temperature profile

With $T_{\text{surf}} \approx 3,500$ K and an adiabatic gradient $\mathrm{d}T/\mathrm{d}r \approx -0.3\ \mathrm{K\,km^{-1}}$, the central temperature reaches $\approx 5,000$ K. The peridotite solidus at 135 GPa is about $4,300$ K (Stixrude & Lithgow-Bertelloni, 2014) : melting is thermodynamically stable throughout the mantle, confirming Hypothesis H1.

B Demonstration of the $b' = 1$ attractor

The parameter b' is defined as the ratio between the critical ejection speed U_{crit} and the geostrophic equilibrium speed U_{eq} of the CMT :

$$b' = \frac{U_{\text{crit}}}{U_{\text{eq}}} = \frac{2\Omega \sin\varepsilon \sqrt{2|g_{\text{eff}}| R_{\text{eq,proto}}}}{|g_{\text{eff}}|}. \tag{62}$$

Evolution due to Fe-Ni segregation. The CMT density decreases as $\rho_{\text{CMT}}(t) = \rho_0(1 - \xi_{\text{Fe}}(t))$, where $\xi_{\text{Fe}}(t)$ is the fraction of Fe-Ni segregated. The effective gravity $|g_{\text{eff}}|$ is proportional to ρ_{CMT}, so

$$\left.\frac{\mathrm{d}b'}{\mathrm{d}t}\right|_{\text{segregation}} = \frac{1}{2}\frac{\dot{\rho_{\text{CMT}}}}{\rho_{\text{CMT}}}\, b' > 0, \tag{63}$$

since $\dot{\rho_{\text{CMT}}} < 0$: Fe-Ni segregation increases b'.

Evolution due to ejection. An ejection carries away a fraction $\epsilon = M_{\text{ej}}^{\max}/M_{\text{CMT}} \approx 0.1$ of the CMT mass. Angular momentum decreases according to $\Delta\Omega/\Omega \approx -\epsilon$, and effective gravity decreases according to $\Delta|g_{\text{eff}}|/|g_{\text{eff}}| \approx -\epsilon$ (proportional to the mass lost). Thus,

$$\left.\frac{\Delta b'}{b'}\right|_{\text{ejection}} = \frac{\Delta\Omega}{\Omega} - \frac{1}{2}\frac{\Delta|g_{\text{eff}}|}{|g_{\text{eff}}|} \approx -\epsilon + \frac{1}{2}\epsilon = -\frac{1}{2}\epsilon < 0 : \tag{64}$$

ejection decreases b'.

Stable fixed point. The two trends oppose each other : segregation increases b', ejection decreases it. The equilibrium point $b' = 1$ is a stable attractor. For $b' < 1$, the system is unstable and an ejection occurs, which further decreases b' (since $\Delta b' < 0$) ; for $b' > 1$, the system is stable, no ejection occurs, and segregation pushes b' upward, toward 1, from below. The point $b' = 1$ is therefore a stable attractor of the coupled segregation–ejection system.

C Complete derivation of the λ coefficient

This appendix gives the complete analytical derivation of the stability coefficient $\lambda(U)$, with dimensional verification and, for each term, its physical history. It follows the Solberg-Høiland parcel displacement formalism (Solberg, 1936; Høiland, 1941; Tassoul, 1978) adapted to the TPT geometry, and leads to the form used in Section 9.2.

C.1 The stability equation

The idea of the method fits in one sentence : one mentally displaces a small fluid parcel along the radius, sums the forces that bring it back or carry it away, and the sign of the sum tells whether the torus is confined or ejected. For an infinitesimal radial displacement ξ_r of a CMT parcel :

$$\ddot{\xi}_r = \lambda(U)\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{65}$$

The displacement is *spherical* radial and the latitude is measured from the instantaneous dynamical equator of the body — no orbital reference (ecliptic) enters the internal dynamics, which knows only Ω and the local vertical. The angle ε_{dyn} denotes the *colatitude of the CMT* relative to this dynamical equator : $\sin\varepsilon_{\text{dyn}} = \cos\phi$, where ϕ is the latitude of the torus ($\varepsilon_{\text{dyn}} = 70^\circ \leftrightarrow \phi = 20^\circ$, at the heart of the intertropical belt).

C.2 The Rayleigh (epicyclic) term

The displaced parcel conserves its specific angular momentum $L = rU$, while the background flow follows the profile $U(r) \propto r^\alpha$. Finding itself at a radius where the background rotates differently, the parcel experiences a restoring force : this is the Rayleigh mechanism (Rayleigh, 1917), whose intensity is the square of the epicyclic frequency,

$$\kappa^2 = \frac{1}{r^3}\frac{\mathrm{d}\,(rU)^2}{\mathrm{d}r} = \frac{2(1+\alpha)\,U^2}{r^2}, \tag{66}$$

obtained by differentiating $(rU)^2 \propto r^{2(1+\alpha)}$. This term is always positive (stabilizing) for any physical profile ($\alpha > -1$). Essential point : this centrifugal charge gradient contains *by itself* the entire centrifugal effect. Adding a separate "centrifugal pressure" term would amount to counting the same effect twice ; the present derivation contains none.

C.3 The non-traditional Coriolis component (Eötvös effect)

In a rotating frame, any prograde flow is *lightened* radially : the Coriolis force applied to the azimuthal speed U has a component along the local vertical, $+2\Omega U\cos\phi$, directed outward. This is the Eötvös effect — the same one that makes an object lighter when it moves eastward. The "traditional approximation" of geophysics neglects this component ; it must be retained at low latitudes and for deep flows (Gerkema et al., 2008) — precisely the regime of the intertropical belt. Writing

$\sin \varepsilon_{\mathrm{dyn}} = \cos \phi$:

$$a_{\text{Eötvös}} = +\frac{2\Omega \sin \varepsilon_{\mathrm{dyn}}}{r}\, U\, \xi_r, \tag{67}$$

destabilizing term : it opposes gravitational confinement.

C.4 The role of Euler acceleration (nutation)

The rotation axis nutates, and one might be tempted to add to λ an Euler term from $-\dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$. One must resist this temptation, for a structural reason : λ gathers accelerations *proportional to the displacement* ξ_r, whereas the Euler acceleration is a *forcing* dependent on time, independent of ξ_r (and a combination such as $|\dot{\Omega}|\, r/\Omega^2$ is moreover homogeneous to meters, not s^{-2}). Its physical place is elsewhere, and it is already occupied in the TPT : nutation slowly modulates the effective colatitude $\varepsilon_{\mathrm{dyn}}(t)$ — the modulation parameter $|\dot{\Omega}|/\Omega^2 \approx 0.08 \sin \theta_{\mathrm{nut}} \ll 1$ guarantees the adiabatic regime — and periodically lowers Ω_{crit} through the Eötvös term $2 \sin \varepsilon_{\mathrm{dyn}}$. This is precisely the *parametric amplification* channel of Section 9.2 : the Euler effect is real, maximal at nutation peaks, and enters the mechanism as a modulation of the threshold, not as a term in λ.

C.5 Buoyancy : thermal–compositional split

The density gradient of the body has two sources, and the derivation splits them in the manner of the Ledoux criterion. The *thermal* part — the magma is cooled from above, hence superadiabatic — enters through the Brunt–Väisälä frequency (Väisälä, 1925; Brunt, 1927) along gravity :

$$-N^2 > 0 \qquad (N^2 < 0,\ \text{convection}), \tag{68}$$

destabilizing contribution that sustains convection. The *compositional* part — Fe-Ni segregation makes the body denser at depth — is measured by $\beta \equiv -\mathrm{d} \ln \rho / \mathrm{d} \ln r > 0$ and enters through the *centrifugal* channel. The picture is simple : the centrifugal force is a gravity that points outward ; a stratification stable for true gravity (dense at the bottom) is, for this inverted gravity, a Rayleigh-Taylor-type configuration, which pushes the dense fluid outward :

$$a_\beta = +\frac{\beta\, U^2}{r^2}\, \xi_r. \tag{69}$$

Destabilizing term — and, as will be seen, indispensable : it is the engine's fuel. Fe-Ni segregation thus does double duty in the TPT : it accelerates rotation by contraction (ice-skater effect) and it creates the stratification whose centrifugal buoyancy fuels the ejection.

C.6 Assembled coefficient and dimensional verification

Summing the confining gravity $-g_{\text{eff}}/r$, thermal buoyancy $-N^2$, the Eötvös effect (67), the Rayleigh restoring force (66) and the centrifugal buoyancy (69) :

$$\lambda(U) = -\frac{g_{\text{eff}}}{r} - N^2 + \frac{2\Omega\sin\varepsilon_{\text{dyn}}}{r}U - \frac{2(1+\alpha)-\beta}{r^2}U^2. \tag{70}$$

Each term is in s^{-2} : $[g_{\text{eff}}/r] = (\text{m}\,\text{s}^{-2})/\text{m} = \text{s}^{-2}$; $[N^2] = \text{s}^{-2}$; $[\Omega\sin\varepsilon_{\text{dyn}}\,U/r] = \text{s}^{-1}(\text{m}\,\text{s}^{-1})/\text{m} = \text{s}^{-2}$; $[U^2/r^2] = (\text{m}\,\text{s}^{-1})^2/\text{m}^2 = \text{s}^{-2}$.

C.7 Why is the Eötvös term destabilizing here ?

A legitimate objection must be addressed head-on. In the canonical derivation in cylindrical coordinates — displacement in the rotation plane, conserved absolute angular momentum $j = r(U + \Omega r\sin\varepsilon_{\text{dyn}})$ — the Coriolis contribution is absorbed into the epicyclic frequency and becomes *stabilizing*. This absorption is exact at the strict equator. But the CMT is not an equatorial line : it is a belt whose center of mass is off the equator, and three established results change the picture there. (1) Off the equator, the epicyclic absorption acts only on the cylindrical projection of the displacement and the non-traditional component must be retained (Gerkema et al., 2008). (2) More importantly, the TPT rotation varies on spheres ($U(r)$ in spherical radius) and not on cylinders : it violates the second Goldreich–Schubert stability condition ($\partial\Omega/\partial z = 0$) (Goldreich & Schubert, 1967), so that there exist *inclined* displacement paths along which the Solberg-Høiland criteria are violated — non-diffusive instability as soon as $N^2 \leq 0$, which is the case here, *independently* of the sign attributed to the radial Coriolis term. (3) The classical non-viscous criteria moreover *underestimate* instability as soon as the Prandtl number deviates from unity (McIntyre, 1970), which is the case for silicate magma. Finally, the latitudinal shear at the torus edge, $\partial U/\partial y \sim U/W$, is comparable to 2Ω (ratio ≈ 0.95 at nominal values) : the CMT operates near the equatorial inertial instability criterion (Dunkerton, 1981). The geometric condition of the present criterion — unfavorable absolute angular momentum gradient along inclined paths at the CMT edge — is verifiable by 3D SPH simulation and joins Limitation L14.

C.8 The rotation threshold

Setting $\lambda = 0$ and substituting $U = \Omega r$:

$$\Omega_{\text{crit}}^2 = \frac{g_{\text{eff}}/r + N^2}{2\sin\varepsilon_{\text{dyn}} - 2(1+\alpha) + \beta}. \tag{71}$$

A finite threshold exists only if the denominator D is positive, i.e., $\beta > 2(1+\alpha) - 2\sin\varepsilon_{\text{dyn}}$. The driving role of centrifugal buoyancy is read here : at nominal values, $2\sin\varepsilon_{\text{dyn}} - 2(1+\alpha) = 1.879 - 3 = -1.12 < 0$ — without β, no rotation would ever trigger ejection. With $\varepsilon_{\text{dyn}} = 70^\circ$ at the nutation peak, $\alpha = 0.5$, $\beta = 2.5$ and $N^2 = -1.93 \times 10^{-7}\ \text{s}^{-2}$, the denominator is $D = 1.379$ and the threshold $\Omega_{\text{crit}} \approx 1.27\,\Omega_0$. Note that the present form, freed from double counting, halves the sensitivity of the threshold to the velocity profile ($\partial D/\partial\alpha = -2$ instead of -4), and coincides numerically with the earlier form at $\alpha = 0.5$. This threshold is a well-defined surface in parameter space $(\Omega, \varepsilon_{\text{dyn}}, \beta)$, giving the mechanism a predictive character : ejection occurs when $\Omega > \Omega_{\text{crit}}$, the crossing occurring by internal means at conserved angular momentum — segregation contraction (ice-skater effect) and parametric amplification (Section 9.2) — the Late Veneer impulsive trigger acting only as modulation at the nutation peak.

D Demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ criterion

The Bingham-Kelvin-Helmholtz number compares the yield stress to the KH shear stress :

$$\mathrm{Bi}_{\mathrm{KH}} = \frac{\tau_y\, k}{\rho\, \sigma_{\mathrm{KH}}\, \Delta U}, \tag{72}$$

where k is a geometric factor (~ 1), $\sigma_{\mathrm{KH}} \sim \Delta U/\lambda$ is the KH growth rate, and λ is the wavelength of the perturbation. Thus,

$$\mathrm{Bi}_{\mathrm{KH}} \sim \frac{\tau_y\, \lambda}{\rho\, \Delta U^2}. \tag{73}$$

For the CMT : $\tau_y \sim 10^2$–10^4 Pa, $\lambda \sim R_{\text{torus}} \sim 10^7$ m, $\rho \sim 3,000\ \text{kg}\,\text{m}^{-3}$, $\Delta U \sim U_{\text{crit}} \sim 10^4\ \text{m}\,\text{s}^{-1}$. One obtains $\mathrm{Bi}_{\mathrm{KH}} \sim 3 \times 10^{-3}$–$3 \times 10^{-1}$.

Effective cohesion is the product of three independent reinforcements (atmospheric confinement, internal crystal network, Rayleigh-Taylor suppression), detailed in Section 9.1 :

$$\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} = \mathrm{Bi}_{\mathrm{KH}} \times f_{\text{atm}} \times f_{\text{crist}} \times f_{\text{RT}}, \tag{74}$$

with $f_{\text{atm}} \sim 10$–100 (atmospheric confinement, $P \sim 10^5$–10^7 Pa), $f_{\text{crist}} \sim 3$–10 (internal crystal network), and $f_{\text{RT}} \sim 10$–50 (Rayleigh-Taylor suppression, $\rho_{\text{flow}}/\rho_{\text{atm}} \sim 200$). With nominal values, $\mathrm{Bi}_{\mathrm{KH}} \sim 3 \times 10^{-2}$ and the product of the three reinforcements is ~ 300, giving :

$$\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} \sim 9, \tag{75}$$

well above 1. Cohesion is therefore ensured.

Recognized limitation : this calculation is a nominal estimate. A 3D SPH simulation with full BH rheology is required to validate the exact values, in particular the nonlinear interaction between the three effects (Limitation L14).

E Derivation of the Elsasser number and B_{crit}

Balancing the Coriolis force $F_C \sim 2\rho\Omega U$ and the Lorentz force $F_L \sim B^2/(\mu L)$, with $\eta \sim UL$ (magnetic Reynolds number of order unity at the dynamo threshold) :

$$B^2 \sim 2\mu\rho\Omega\eta, \qquad \Rightarrow \qquad \Lambda \equiv \frac{B^2}{\rho\mu\eta\Omega} \sim 2, \tag{76}$$

i.e., $\Lambda \sim \mathcal{O}(1)$, consistent with values $\Lambda \sim 0.1$–1 inferred for the present Earth's core (Christensen & Aubert, 2006). Using $\eta = (\mu_0\sigma)^{-1}$ with $\sigma \approx 1.5 \times 10^6\ \mathrm{S\,m^{-1}}$, a representative electrical conductivity of liquid iron at core conditions (Pozzo et al., 2012), $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$, $\rho \approx 10^4\ \mathrm{kg\,m^{-3}}$, $\Omega \approx 4.99 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, one obtains $B_{\mathrm{crit}} \approx 2.6$ mT, of the same order of magnitude as the field inferred within the present Earth's core (Christensen & Aubert, 2006). Note that this value assumes Ω at its formation value ($T = 3.5$ h). At dynamo onset ($t \approx 330$–350 Ma), after drainage of angular momentum by the evection resonance (Section 11.5) and ordinary tidal braking, Ω is reduced by a factor ≈ 2; since $B_{\mathrm{crit}} \propto \sqrt{\Omega}$, one then obtains $B_{\mathrm{crit}} \approx 1.8$–$1.9$ mT — same order of magnitude, conclusion unchanged.

F Stokes sedimentation timescale of Fe-Ni

Stokes sedimentation speed for an Fe-Ni droplet of radius $r \approx 5$ cm in molten silicate :

$$v_{\mathrm{Stokes}} = \frac{2}{9}\frac{(\rho_{\mathrm{Fe}} - \rho_{\mathrm{sil}})\, g_{\mathrm{eff}}^{(0)}\, r^2}{\eta} \approx \frac{2}{9}\frac{3,300 \times 6.42 \times (0.05)^2}{0.1} \approx 118\,\mathrm{m\,s^{-1}}. \tag{77}$$

Over a characteristic sedimentation distance $\ell \approx 30$ km : $\tau_{\mathrm{Stokes}} = \ell / v_{\mathrm{Stokes}} \approx 254$ s. Robust for $r \in [1, 10]$ cm and $\eta \in [0.05, 1]$ Pa s.

G Accretion of magmatic ejecta : why perfect cohesion is not required

A recurring objection to the TPT mechanism concerns the cohesion of the CMT flow during transit between the proto-Earth's surface and the Roche radius. This appendix demonstrates that, even without perfect cohesion of the flow, the physical properties of the ejected magma and orbital mechanics guarantee accretion into a single body.

The magma is not dust. The ejected material is a silicate magma at $T \approx 3,000$–$4,000$ K. Its dynamic viscosity is $\eta \approx 1$–100 Pa s (Giordano et al., 2008) — comparable to warm honey. Unlike solid rock fragments or a gas cloud, a high- temperature magma does not disperse : it deforms and stretches, but remains dense and cohesive at the macroscopic scale. Two sticky magma drops at low relative velocity do not bounce off each other — they fuse. Viscous coalescence replaces mechanical cohesion as the dominant aggregation mechanism once the flow is fragmented.

Orbital concentration. The ejecta from a given TPT episode share, to first order, the same initial orbital energy budget. By conservation of energy and angular momentum, they populate a band of nearby orbits rather than dispersing isotropically over 4π steradians. This orbital concentration is fundamentally different from an isotropic explosion : it is the same physical principle that produces accretion disks from tidally disrupted bodies. The toroidal geometry of the CMT reinforces this effect : ejection occurs preferentially within the oblique intertropical belt $|\phi| < 30°$, concentrating ejecta within a limited azimuthal solid angle. The resulting debris field is geometrically thin, maximizing mutual encounter rates.

Beyond the Roche radius : accretion is inevitable. Beyond the Roche radius $R_{\text{Roche}} = 2.456 R_{\oplus}$, the self-gravity of the ejecta cloud overcomes tidal disruption. For co-orbital magma fragments of total mass $M_{\text{cloud}} \approx 2\text{–}4 \times 10^{22}$ kg, the gravitational accretion timescale is :

$$\tau_{\text{acc}} \approx \frac{1}{\sqrt{G\rho_{\text{cloud}}}} \approx 10^3\text{–}10^5\ \text{yr}, \tag{78}$$

much shorter than the thermal lifetime of the magma ($\tau_{\text{cool}} \sim 10^6\text{–}10^7$ yr for a Moon-sized body, Elkins-Tanton 2008). During this accretion window, the magma remains above its solidus : every collision is plastic, not elastic. There is no fragmentation cascade, no loss by rebound. The capture efficiency f_{cap} therefore has a physical lower bound independent of the BH cohesion assumption :

$$f_{\text{cap}}^{\min} \approx \frac{M_{\text{cloud}}(r > R_{\text{Roche}})}{M_{\text{ej}}^{\text{total}}} \gtrsim 0.45\text{–}0.55, \tag{79}$$

the fraction lost ($\approx$ 45–55%) corresponding to ejecta that fall back onto the proto-Earth or escape on hyperbolic orbits.

Implication for the theory. The capture efficiency $f_{\text{cap}} = 0.70$ adopted in the main text is therefore not a free parameter that the theory must defend at all costs. Even in the pessimistic scenario where the CMT flow fragments completely upon ejection, orbital mechanics and magma viscosity guarantee the accretion of the co-orbital debris cloud beyond the Roche radius. The critical question shifts from "does the flow remain cohesive ?" to "what fraction of ejecta reaches the Roche radius ?" — a question that depends on the velocity distribution of ejecta, and not on BH rheology. This reformulation makes the theory more robust : cohesion is advantageous, but not necessary.

H Complementary mechanisms for the ejection threshold

This appendix identifies the physical mechanisms that contribute to reaching the ejection threshold $\lambda(U) = 0$. The threshold is a condition on Ω :

$$\Omega > \Omega_{\text{crit}} \approx \sqrt{\frac{|g_{\text{eff}}|}{2\sin\varepsilon\, r}} \cdot \sqrt{\frac{1+\alpha}{r}}. \tag{80}$$

With the nominal proto-Earth parameters ($\Omega_0 = 4.99 \times 10^{-4}$ rad/s, $\varepsilon = 55°$), the threshold is $\Omega_{\text{crit}} \approx \sqrt{2}\,\Omega_0$.

The angular momentum input required to reach this threshold is provided by continuous planetesimal bombardment. The late accretion (the "Late Veneer") is delivered by a small number of large bodies (radius $\gtrsim$ 750 km) (Raymond et al., 2013). Geodynamic constraints on the supply of highly siderophile elements to the mantle — impactors larger than $\sim$ 1 km deliver metal that sinks to the core rather than remaining entrained in the convective mantle — place the total late accretion mass up to $\sim$ 5% of the Earth's mass (Anslow et al., 2026). Assigning to this mass a characteristic impact speed $v_{\text{imp}} \sim 10$ km s^{-1} and a lever arm of order r, the cumulative available angular momentum is :

$$L_{\text{imp}} \sim M_{\text{imp}}\, v_{\text{imp}}\, r \approx 2.2 \times 10^{34}\ \text{J s}. \tag{81}$$

This input is sufficient to raise Ω from Ω_0 to Ω_{crit} in the geometrically most favorable case ($\varepsilon \approx 90°$), and provides the necessary margin for ejection to occur at the peaks of the nutation cycle. The Euler oscillation term λ_4 modulates the effective obliquity ε, but does not provide the main driving force.

I Persistence of post-ejection oscillation and repeated opportunities

The ejection threshold is not a single-pass condition. After an ejection, Ω falls below Ω_{crit}, but the system does not stop immediately. The cavity left by the ejection is bounded by the confining atmosphere and by the residual *'a'ā* crust of the CMT. These boundaries are reflective rather than absorbent (as established by the $\text{Bi}_{\text{KH}} \gg 1$ criterion, Appendix D), so that the post-ejection nutation persists over several cycles.

The obliquity excursion envelope $\varepsilon_{\text{exc}}(t)$ follows a two-time regime :

$$\varepsilon_{\text{exc}}(t) \approx \begin{cases} \varepsilon_{\text{exc}}^{(0)} \cos(\omega_{\text{nut}}^{(\text{post})} t) & 0 \le t \lesssim t_{\text{rec}} \\ \varepsilon_{\text{exc}}^{(0)}\, e^{-(t-t_{\text{rec}})/\tau_{\text{BH}}} \cos(\omega_{\text{nut}}^{(\text{post})} t) & t \gtrsim t_{\text{rec}} \end{cases} \tag{82}$$

where $\omega_{\text{nut}}^{(\text{post})} = 2\pi / T_{\text{nut}}^{(\text{post})}$, with $T_{\text{nut}}^{(\text{post})} \approx 30$ h, and $\tau_{\text{BH}} \sim 10^2$–$10^4$ s (Eq. 1) for the decay that takes over once the plateau is complete.

Over the recharge window duration, $t_{\text{rec}} \approx 10$–$12$ days, this plateau covers :

$$N_{\text{opp}}^{(\Delta)} = \frac{t_{\text{rec}}}{T_{\text{nut}}^{(\text{post})}} \approx \frac{240\text{–}288\,\text{h}}{30\,\text{h}} \approx 8\text{–}10 \tag{83}$$

complete cycles, each bringing ε transiently close to its post-shock maximum excursion. The ejection threshold is therefore not a single attempt but a modest number of repeated opportunities, each with a high individual probability.

Implication for the ejection mechanism. The post-ejection plateau window is a period of ≈ 8–10 nutation cycles (about 10–12 days) during which ε remains close to its post-shock maximum. A major Late Veneer impact occurring during this window benefits from a double advantage :

1. ε is already close to its maximum, so Ω_{crit} is minimal ;
2. The impulsive torque of the impact provides a jump $\Delta\Omega$ added to Ω already near the threshold.

The combination of the two erases the threshold and triggers the next ejection. This mechanism links the ejection episodes to late impacts (Late Veneer), and not to a simple continuous accumulation of angular momentum. The isotopic signature of these late impacts (excess of siderophiles) should be present in the outermost lunar layers (Prediction P22).

J List of abbreviations

BH Bingham-Herschel, rheological law.

BiKH Bingham-Kelvin-Helmholtz number.

CME Coronal Mass Ejection.

CMT Coherent Magmatic Torus.

ε_{dyn} Colatitude of the CMT relative to the instantaneous dynamical equator ($\sin \varepsilon_{\text{dyn}} = \cos \phi$).

ε_{orb} Spin-orbit angle (orbital obliquity).

FSS Farside Seismic Suite.

Hf-W Hafnium-tungsten chronometer.

LHB Late Heavy Bombardment (≈ 3.9 Ga).

LEMS Lunar Environment Monitoring Station.

LMO Lunar Magma Ocean.

NLS Nonlinear Schrödinger equation.

POB Proto-Orbital Body.

SPA South Pole–Aitken (basin).

SPH Smoothed Particle Hydrodynamics.

TPT Triple Phase Transition.

Declarations

Data and materials availability. All data generated or analyzed during this study are included in this published article (and its appendices). The interactive simulation accompanying this work is freely accessible at https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html.

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